

Nonintegral Laplacian spectra of ideal intersection graphs

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Abstract

We study Laplacian integrality for the graph of nonzero proper ideals of $\mathbb{Z}_n$, where distinct ideals are adjacent when their intersection is nonzero. For squarefree composite n , the graph is Laplacian integral exactly when n has two prime factors. For three distinct primes, we prove nonintegrality for every repeated-exponent triple (a, a, b) with $a, b \geq 1$ and every exponent triple with minimum entry at most seven. For ordered $2 \leq a \leq b \leq c$, we also prove nonintegrality whenever $c \geq 4a^2 - 2a$. A symmetric Schur complement supplies a positive quotient root in $(0, 3)$ for minimum exponent at least four and proves further nonintegral families, including every exponent span at most three and an unbounded balanced region. Arithmetic arguments settle every common exponent divisor at least three, every all-odd triple, and every common 2-adic valuation. Shifted root distributions settle every permutation of $(3, 3, 2)$ and $(3, 0, 0)$ modulo four. A combined mixed-parity theorem includes these exclusions and, for residue roles $(1, 3, 2)$ modulo four, combines the sum congruence $a + c \equiv b(b + 2) \pmod{128}$, a derivative restriction on b , and a binary-cubic parity condition. For minimum at least four, it also requires the endpoint-two equation and a linear bound on the maximum exponent. The same linear bound applies to all-even triples and mixed pattern $(3, 3, 0)$ modulo four. The repeated-exponent proof combines discriminant arithmetic with a consecutive-integer interval for a symmetric cubic. Exact finite certificates exhaust the exceptional divisor levels, the minimum-three region and the cutoff regions at minima four through seven; an endpoint certificate also settles every triple with second-smallest exponent at most fifteen. On the endpoint-one surface at real minimum exponent $a \geq 4$, we prove that one is a simple smallest positive quotient root and all four other roots exceed a ; their smallest lies below both c and $a + b$. The remaining unequal-exponent and nonsquarefree higher-prime cases stay open.

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1 Introduction

The ideal intersection graph associates a graph to the intersection structure of the nonzero proper ideals of a ring [8]. For $\mathbb{Z}_n$, its isomorphism type depends on the exponents in the prime factorization of n . We ask which of these graphs have integer Laplacian eigenvalues.

Graphs defined by ring multiplication provide related constructions: Beck's coloring graph and the zero-divisor graph [3, 2], and the annihilating-ideal graph [4, 5, 1]. Their multiplication-based adjacency differs from the nonzero-intersection convention used here.

The Laplacian viewpoint originates in algebraic connectivity [10] and its spectral development [13, 14]; general graph-spectral background is given in [9, 6]. Equitable partitions and generalized joins [12, 7] reduce the present graphs to small invariant matrices.

Nikandish [15] develops a generalized-join formula for the Laplacian spectrum and a Schur–Weyl reduction for equal prime exponents. His results include nonintegrality for $(pqr)^k$ for every $k \geq 1$. His Question (Q3) asks for the full integrality characterization; the present arguments address this question for additional exponent families.

We prove a complete classification for squarefree composite integers (Theorem 3.1), an explicit quadratic obstruction for $(1, 1, k)$ (Theorem 4.1), and nonintegrality for every repeated-exponent triple (a, a, b) , $a, b \geq 1$ (Theorem 7.1). The last result combines an arithmetic restriction on the antisymmetric block with a consecutive-integer sign interval for the symmetric cubic. The boundary family $b = (a - 1)(2a - 1)$ is proved in Theorem 5.1. An exact divisor criterion isolates every possible square discriminant for fixed a . A second divisor criterion fixes the smaller-factor index instead, giving a finite reduction over all a (Proposition 6.6). The first three nonboundary factor levels have complete finite modular certificates. Every later level lies strictly below the uniform sign threshold, closing the whole repeated-exponent family.

For a unit exponent, the complement of the general six-support quotient has a nonzero eigenvalue strictly between 0 and 1, except for three small pairs already covered by $(1, 1, k)$. This proves nonintegrality for every $(1, b, c)$ (Theorem 8.1), including fully unequal triples. Positive-coefficient expansions at a minimum exponent of two, together with one exceptional rational sign interval, then settle every $(2, b, c)$ (Theorem 9.1). Thus every three-prime exponent vector with minimum entry at most two is covered. The same endpoint polynomial gives a uniform cutoff $c \geq 4a^2 - 2a$ for ordered $2 \leq a \leq b \leq c$ (Proposition 10.1). For a minimum exponent of three, this leaves exactly 325 pairs. An exhaustive integer sign certificate, positive second-endpoint expansions and one rational exception settle all of them (Theorem 10.2). Thus every triple with minimum entry at most three is nonintegral.

For nearby unequal exponents, low roots can occur in pairs without an endpoint sign change. A symmetric Schur complement gives an inertia criterion for two roots in a specified interval (Proposition 11.1). At the upper endpoint three, four bounded-gap families satisfy the criterion for every $a \geq 4$. Positive first-endpoint factors and root-free second-endpoint factors exclude both possible integers. This proves Theorem 11.2, including the full family $(a, a + 1, a + 2)$ and every triple whose maximum minus minimum is at most three.

The requested diagnostic for minimum entries four through seven exhausts the finite region left by the uniform cutoff. All 27 562 quintics have positive nonsquare discriminants, verified by independent integer Sylvester determinants. Together with the written tail and repeated-exponent proofs, this yields Corollary 11.4: every triple with minimum entry at most seven is nonintegral. The dependence on the complete finite certificate is explicit.

The Schur complement also admits a uniform comparison that covers mixed diagonal signs. For every triple with minimum exponent at least four, the complement quotient has a positive eigenvalue below three (Theorem 12.2). If $(a - 2)(a + b + c - 2) > 2bc$, that eigenvalue lies in $(2, 3)$ (Corollary 12.3). This settles the growing balanced region where the minimum exponent is at least three times the exponent span plus eight (Corollary 12.4). The universal bound also reduces any possible integral triple in this range to the endpoint-zero condition $h_C(1)h_C(2) = 0$.

An integer-scaling argument then settles every triple with common exponent divisor at least three (Theorem 13.1). Finite-field factorization settles every all-odd triple and explicit common-residue classes, including residues 1, 3, 4 modulo five (Theorem 13.2). These arithmetic arguments allow unbounded exponent differences and do not require excluding all integer quotient roots. Scaling also settles every common 2-adic valuation (Corollary 13.4). Theorem 13.5 collects the mixed-parity exclusions and necessary conditions: six classes modulo eight, the full $(3, 3, 2)$ and $(3, 0, 0)$ classes modulo four, and the sum, derivative and binary-cubic restrictions for the $(1, 3, 2)$ role. Its low-root consequences confine $(1, 3, 2)$ and $(3, 3, 0)$ to the endpoint-two surface inside a

linear region. Theorem 14.1 combines that linear bound, the endpoint divisor constraints, the complete middle-exponent-through-fifteen result and two further modulo-three exclusions. One worked shifted-root example is given in the main text; all case calculations and endpoint details are grouped in Appendices B and C.

On the endpoint-one surface, Theorem 14.2 proves a sharp exponent-sum lower bound and places the four residual roots above the minimum exponent. Principal interlacing bounds their smallest below $\min\{c, a + b\}$. The resulting constant-term divisor window in Corollary 14.3 focuses the remaining integer-splitting question. Its proof uses a twelve-term positive identity and a determinant-sign count, with no finite base.

The graph and lifting arguments are given explicitly, so the proofs do not require a complete spectral decomposition. Appendix A describes the exact computations supporting the displayed formulas and finite certificates. The classification for arbitrary exponent vectors remains open.

2 The graph and its invariant subspaces

Write $n = \prod_{i=1}^t p_i^{k_i}$, where the primes are distinct and $k_i \geq 1$. Denote by G_n the graph whose vertices are the nonzero proper ideals of $\mathbb{Z}_n$, with distinct ideals adjacent if their intersection is nonzero. We use the combinatorial Laplacian $L = D - A$; the graph is *Laplacian integral* when every eigenvalue of L is an integer, including when the graph is disconnected.

The ideal $(\prod_i p_i^{\alpha_i})$ is encoded by $0 \leq \alpha_i \leq k_i$. We exclude $(0, \dots, 0)$, which encodes the whole ring, and $(k_1, \dots, k_t)$, which encodes the zero ideal. Ideal intersection is generated by the least common multiple. Therefore two distinct vertices α, β are adjacent exactly when

$$\max(\alpha_i, \beta_i) < k_i \quad \text{for some } i.$$

Equivalently, putting $S(\alpha) = \{i : \alpha_i < k_i\}$, adjacency is $S(\alpha) \cap S(\beta) \neq \emptyset$.

For nonempty $S \subseteq [t]$, the support class $\mathcal{C}_S$ is a clique of size $\prod_{i \in S} k_i$, except that the full support has size $u = \prod_i k_i - 1$. Edges between distinct support classes are complete or absent according as their supports intersect or are disjoint. The full support consists of universal vertices. Remove them and call the resulting graph H ; thus $G_n = K_u \vee H$, with the evident interpretation for $u = 0$.

Lemma 2.1 (Lifting across the universal vertices). *If $L(H)v = \lambda v$ and $\lambda \neq 0$, extension of v by zero on the u universal vertices is an eigenvector of $L(G_n)$ with eigenvalue $\lambda + u$.*

Proof. Since $L(H)$ is symmetric with zero row sums, $\lambda \neq 0$ implies $\sum v = 0$. At a vertex of H , its degree increases by u and all new neighbor values are zero. At a universal vertex the Laplacian action is $-\sum v = 0$. These are exactly the claimed eigenvector equations. $\square$

For $(k_1, k_2, k_3) = (a, a, b)$, H has six support classes of sizes a, a, b, a^2, ab, ab on $1, 2, 3, 12, 13, 23$, respectively. The partition into

$$X = \mathcal{C}_1 \cup \mathcal{C}_2, \quad Y = \mathcal{C}_3, \quad Z = \mathcal{C}_{12}, \quad W = \mathcal{C}_{13} \cup \mathcal{C}_{23}$$

is equitable, and on functions constant on these cells the Laplacian acts by

$$B = \begin{pmatrix} a^2 + ab & 0 & -a^2 & -ab \\ 0 & 2ab & 0 & -2ab \\ -2a & 0 & 2a + 2ab & -2ab \\ -a & -b & -a^2 & a^2 + a + b \end{pmatrix}. \tag{1}$$

For example, an X vertex has a^2 neighbors in Z and ab in W , while a W vertex has a, b, a^2 neighbors in X, Y, Z . Internal neighbors cancel for cell-constant functions. The cell sizes are $2a, b, a^2, 2ab$, and multiplying B on the left by their diagonal matrix gives a symmetric matrix. Thus B is similar to a real symmetric matrix.

The functions taking values $s, -s$ on $\mathcal{C}_1, \mathcal{C}_2$, $t, -t$ on $\mathcal{C}_{13}, \mathcal{C}_{23}$, and zero elsewhere form a second invariant subspace, with matrix

$$B_- = \begin{pmatrix} a^2 + ab & -ab \\ -a & a^2 + a + b + 2ab \end{pmatrix}. \quad (2)$$

These functions have sum zero. Their eigenvalues therefore also lift by $u = a^2b - 1$, directly by the proof of Lemma 2.1.

Expanding (1) gives $\det(xI - B) = xf_{a,b}(x)$, where

$$\begin{aligned} f_{a,b}(x) &= x^3 - E_1x^2 + E_2x - E_3, \\ E_1 &= 2a^2 + 5ab + 3a + b, \\ E_2 &= a^4 + 7a^3b + 3a^3 + 8a^2b^2 + 11a^2b + 2a^2 + 3ab^2 + 2ab, \\ E_3 &= 2a^2b(a + b + 1)(a^2 + 2ab + 2a + b). \end{aligned} \quad (3)$$

Its roots are real and nonzero: reality follows from the similarity above, and $E_3 > 0$. Every root λ gives an eigenvalue $\lambda + u$ of G_n . A noninteger root of $f_{a,b}$ therefore proves nonintegrality.

3 Squarefree integers

For squarefree integers the support classes have size one, so the graph is described directly by subsets of the prime factors. Alternating tensors provide invariant subspaces for this subset action [11] and give the complete squarefree classification.

Theorem 3.1. *For squarefree composite n , the graph G_n is Laplacian integral if and only if n has exactly two prime factors.*

Proof. Let t be the number of prime factors. For $t = 2$ the graph consists of the two ideals (p) and (q) , with no edge, so $L = 0$. Assume $t \geq 3$. The vertices are the nonempty proper subsets of $[t]$, adjacent when they intersect. On the tensor space encoding all subsets put

$$E = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \quad F = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad J = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad c = 2^t - 1.$$

The principal submatrix of $\Phi = cI - E^{\otimes t} + F^{\otimes t} - J^{\otimes t}$ on the graph vertices is $L(G_n)$: the diagonal entry at S is $2^t - 2 - 2^{t-|S|}$, and the off-diagonal entry is -1 for intersecting subsets and zero otherwise.

With e_0, e_1 denoting the absence and presence basis, set $w = e_0 \otimes e_1 - e_1 \otimes e_0$. For every 2×2 matrix T , one has $T^{\otimes 2}w = (\det T)w$. A paired factor w thus contributes $2, -1, 0$ under E, F, J , respectively. It vanishes at the empty and full subsets. Each subspace used below is invariant for Φ and is supported on graph vertices, so its restriction is also a restriction of L .

If $t = 2r + 1$, $r \geq 1$, use $w^{\otimes r} \otimes \text{span}\{e_0, e_1\}$. Putting $a = 2^r$ and $s = (-1)^r$, the restricted matrix is

$$\begin{pmatrix} c - 2a + s & s \\ s & c - a \end{pmatrix},$$

with discriminant $(a - s)^2 + 4$. The integer $h = a - s$ is at least 3, so $h^2 < h^2 + 4 < (h + 1)^2$. Both eigenvalues are irrational.

If $t = 2r$, $r \geq 2$, use $r - 1$ paired factors and the symmetric subspace on the last two coordinates, with basis $e_0e_0, (e_0e_1 + e_1e_0)/\sqrt{2}, e_1e_1$. Put $a = 2^{r-1}$ and $s = (-1)^{r-1}$. The restriction is $cI - K$, where

$$K = \begin{pmatrix} 4a - s & -s\sqrt{2} & -s \\ -s\sqrt{2} & 2a - s & 0 \\ -s & 0 & a \end{pmatrix}.$$

The leading principal minors of $K - (a - 1)I$ are

$$3a + 1 - s, \quad (3a + 1 - s)(a + 1 - s) - 2, \quad (3a - s)(a + 1 - s) - 2.$$

They are positive for $a \geq 2$ and $s = \pm 1$: the latter two are at least 10 and 8, respectively. Sylvester's criterion gives $\lambda_{\min}(K) > a - 1$. The principal submatrix of $K - aI$ on coordinates 1, 3 is $\begin{pmatrix} 3a - s & -s \\ -s & 0 \end{pmatrix}$, with determinant -1 . A vector supported on those coordinates has a negative Rayleigh quotient, giving $\lambda_{\min}(K) < a$. Hence $c - \lambda_{\min}(K)$ is an eigenvalue of L strictly between the consecutive integers $c - a$ and $c - a + 1$. $\square$

4 The family pqr^k

With two unit exponents, exchanging their coordinates isolates a two-dimensional invariant subspace. Its discriminant gives explicit irrational eigenvalues for every positive third exponent.

Theorem 4.1. *For distinct primes p, q, r and every $k \geq 1$, the graph G_{pqr^k} has the irrational eigenvalues*

$$\frac{6k + 1 \pm \sqrt{(2k + 2)^2 - 3}}{2}.$$

In particular, every permutation of the exponent vector $(1, 1, k)$ gives a nonintegral Laplacian spectrum.

Proof. The support classes 1, 2 have size 1, and 13, 23 have size k . Consider functions with values $s, -s$ on the first two classes, $t, -t$ on the latter two, and zero elsewhere. A vertex of support 1 has $2k$ neighbors and Laplacian value $2ks - kt$. A vertex of support 13 has degree $4k$; after subtracting the same-class and opposite-class values its Laplacian value is $-s + (4k + 1)t$. The swapped classes give their negatives, and all other classes give zero by cancellation. The restricted matrix of the full graph is therefore

$$B_k = \begin{pmatrix} 2k & -k \\ -1 & 4k + 1 \end{pmatrix}.$$

Its characteristic polynomial is $x^2 - (6k + 1)x + 8k^2 + k$, with discriminant $(2k + 2)^2 - 3$. Since

$$(2k + 1)^2 < (2k + 2)^2 - 3 < (2k + 2)^2$$

for $k \geq 1$, the two roots are irrational. Relabeling prime coordinates preserves the graph. $\square$

5 A boundary family with integral antisymmetric block

The following argument was supplied by Reza Nikandish in the public collaboration discussion. We include it because it is the boundary case needed for the linear cutoff in Section 6.

Theorem 5.1. *For every $a \geq 2$, put $C = (a - 1)(2a - 1)$. For distinct primes p, q, r , the graph $G_{p^a q^a r^C}$ is not Laplacian integral, although the antisymmetric block (2) has integer eigenvalues.*

Proof. Specialize $f_{a,b}$ from (3) to $b = C$. For $a = 2$ it is $x^3 - 47x^2 + 702x - 3312$. Its values at 0, 1, 2, 3, 4 modulo 5 are 3, 4, 2, 3, 3. Thus it has no root in $\mathbb{F}_5$ and is irreducible over $\mathbb{Q}$; its real roots are noninteger. Lemma 2.1 gives noninteger graph eigenvalues.

For $a \geq 3$ put $m = 2a^3 - 2a^2 + 2a - 1$. Expansion gives

$$\begin{aligned} f_{a,C}(m) &= 2(5a^4 - 10a^3 + 7a^2 - 1), \\ f_{a,C}(m - 1) &= -2(a - 1)(a + 2)(2a^4 - 7a^3 + 7a^2 - a - 3). \end{aligned}$$

The first value is positive because its parenthesis is $5a^3(a - 2) + 7a^2 - 1$. Writing $z = a - 3 \geq 0$, the last factor in the second value is $2z^4 + 17z^3 + 52z^2 + 68z + 30 > 0$. Consequently $f_{a,C}(m - 1) < 0 < f_{a,C}(m)$, so a real root lies strictly between $m - 1$ and m . Its integer shift by $u = a^2C - 1$ is noninteger.

The characteristic discriminant of B_- is $D = (a + 1)^2b^2 + 2a(3a + 1)b + a^2$. At $b = C$ it equals $(2a^3 - a^2 + a - 1)^2$, and the two eigenvalues are $4a^3 - 3a^2 + a$ and $2a^3 - 2a^2 + 1$. Thus the obstruction in this family comes from the symmetric cubic. $\square$

6 Finite reduction and a linear cutoff

The repeated-exponent decomposition leaves two possible obstructions: a nonsquare quadratic discriminant and a noninteger root of the symmetric cubic. Exact factor pairs reduce the first exceptional set, while sign intervals settle the large-exponent tail.

Fix $a, b \geq 1$, and put

$$h = a + 1, \quad C = (a - 1)(2a - 1), \quad M = a^3(2a + 1), \quad A = h^2b + a(3a + 1).$$

The characteristic discriminant of (2) is

$$D = h^2b^2 + 2a(3a + 1)b + a^2. \tag{4}$$

If D is not a square, this block has irrational eigenvalues, which remain irrational after the integer shift $a^2b - 1$.

Proposition 6.1 (Exact divisor reduction). *For fixed $a \geq 1$, the positive integers b for which D is a square are exactly those obtained from positive factor pairs $d \leq e$ of M such that*

$$b = \frac{d + e - a(3a + 1)}{h^2} \in \mathbb{Z}_{\geq 1}, \quad s = \frac{e - d}{h} \in \mathbb{Z}_{\geq 0}. \tag{5}$$

In that case $D = s^2$.

Proof. The polynomial identity

$$A^2 - h^2D = 4M \tag{6}$$

holds for arbitrary a, b . If $D = s^2$ with $s \geq 0$, then $A - hs$ and $A + hs$ are positive: their product is positive and their sum is $2A > 0$. They have the same parity; since their product is divisible by 4, both are even. Write $A - hs = 2d$ and $A + hs = 2e$. Then $de = M$, $d + e = A$ and $e - d = hs$, giving (5). Conversely those conditions yield $A = d + e$ and $hs = e - d$. Using $(d + e)^2 - (e - d)^2 = 4de = 4M$ in (6) gives $D = s^2$. $\square$

The divisor set is finite for every fixed a ; this criterion has no truncation in b . It gives first a quadratic bound.

Corollary 6.2. *If D is a square then $b \leq C$. Consequently every $b > C$ gives a nonintegral Laplacian spectrum. For $a = 1$, every $b \geq 1$ is covered.*

Proof. Since $(d-1)(e-1) \geq 0$, $A = d + e \leq M + 1$. The identity

$$M + 1 - a(3a + 1) = h^2 C$$

then gives $b \leq C$. The nonsquare case is covered by the antisymmetric block. At $a = 1$, $C = 0$. $\square$

Lemma 6.3. *Every admissible factor pair in Proposition 6.1 satisfies $d \equiv e \equiv 1 \pmod{h}$.*

Proof. From $e - d = hs$ and $d + e = A$, reduction modulo h gives $e \equiv d$ and $2d \equiv 2$, since $a(3a+1) \equiv 2 \pmod{h}$. Also

$$(d-1)(e-1) = M - A + 1 = h^2(C - b). \quad (7)$$

If h is odd, the congruence $2(d-1) \equiv 0$ gives the assertion. If h is even, the alternative to $d \equiv 1$ is $d-1 \equiv h/2 \pmod{h}$. Then $d-1$ and $e-1$ would both equal $h/2$ times an odd integer. Their product divided by h^2 would be an odd integer divided by 4, contradicting (7). This excludes the alternative for either parity of a . $\square$

Theorem 6.4 (Unified linear cutoff). *Let $a \geq 2$ and*

$$R(a) = \frac{2(a-1)(a-2)}{a+2} = 2a - 10 + \frac{24}{a+2}.$$

For every integer $b > R(a)$ and distinct primes p, q, r , the graph $G_{p^a q^a r^b}$ is not Laplacian integral. Every unresolved repeated-exponent pair therefore has square D and $1 \leq b \leq \lfloor R(a) \rfloor$.

Proof. If D is nonsquare the antisymmetric block suffices. Otherwise use Proposition 6.1. When $d = 1$, $e = M$ and $b = C$; Theorem 5.1 settles this case. If $d > 1$, Lemma 6.3 gives $d \geq T = a + 2$. Since $e \geq d \geq T$,

$$\frac{M}{T} + T - (d + e) = (d - T) \left(\frac{e}{T} - 1 \right) \geq 0.$$

Hence

$$b \leq \frac{M/T + T - a(3a + 1)}{h^2} = \frac{2(a-1)(a-2)}{a+2} = R(a).$$

Thus $b > R(a)$ leaves either a nonsquare discriminant or the already settled boundary case. $\square$

At $(a, b) = (4, 2)$ the square-discriminant pair has $d = 6 = a + 2$, so equality in the factor estimate is attained. The graph is still nonintegral, as shown in the next section.

6.1 Settling the first nonboundary factor

The first possible value after $d = 1$ is $d = a + 2$. Since d divides M , the identity

$$M = (a + 2)(2a^3 - 3a^2 + 6a - 12) + 24$$

forces $a + 2 \mid 24$. With $a \geq 2$ this leaves $a = 2, 4, 6, 10, 22$. Substitution into (5) gives $b = 0, 2, 5, 12, 35$, respectively. Excluding $b = 0$, the complete positive- b list and its cubic certificates are:

(a, b)	$f_{a,b}(x)$	Root-free prime
$(4, 2)$	$x^3 - 86x^2 + 2304x - 18816$	5
$(6, 5)$	$x^3 - 245x^2 + 19266x - 488160$	7
$(10, 12)$	$x^3 - 842x^2 + 230160x - 20534400$	13
$(22, 35)$	$x^3 - 4919x^2 + 7887858x - 4132479120$	17

Each polynomial has no root modulo its indicated prime. Its monic cubic is therefore irreducible over $\mathbb{Q}$, and its real nonzero roots lift to noninteger graph eigenvalues. Thus every possible $d = a + 2$ pair is settled, over all a , by this finite divisor argument.

Theorem 6.5 (Second linear cutoff). *For $a \geq 2$ put*

$$S(a) = \frac{(a-3)(2a-3)}{2a+3} = a - 6 + \frac{27}{2a+3}.$$

For every positive integer $b > S(a)$ and distinct primes p, q, r , $G_{p^a q^a r^b}$ is not Laplacian integral. Every remaining square-discriminant case has $d \geq 2a + 3$ and $b \leq \lfloor S(a) \rfloor$.

Proof. Nonsquare D is covered by the antisymmetric block. If D is square, the cases $d = 1$ and $d = a + 2$ are already settled. Every other admissible factor satisfies $d \geq T = 2a + 3$ by Lemma 6.3. Since $e \geq d \geq T$, the same factor inequality gives

$$b \leq \frac{M/T + T - a(3a+1)}{(a+1)^2} = \frac{(a-3)(2a-3)}{2a+3} = S(a).$$

This proves the claim for $b > S(a)$. □

The new bound is strictly smaller than $R(a)$ for all $a \geq 2$, since

$$R(a) - S(a) = \frac{2a^3 - a^2 - a - 6}{(a+2)(2a+3)} > 0.$$

Writing $z = a - 2 \geq 0$, its numerator is $2z^3 + 11z^2 + 19z + 4$.

6.2 A finite reduction for each factor index

For a nonboundary square-discriminant pair, write $d = 1 + hj$, $j \geq 1$. The following criterion fixes j instead of a .

Proposition 6.6. *Fix $j \geq 1$ and set $K_j = (j+1)^3(j+2)$. Every admissible square-discriminant pair of index j is obtained from a positive divisor d of K_j with $a = (d-1)/j - 1 \in \mathbb{Z}_{\geq 2}$. Set $e = M/d$, $v = (e-1)/(a+1)$, and retain exactly those candidates with $d \leq e$ and $b = C - jv \geq 1$. Then e, v, b are integers and $D = (v-j)^2$. Conversely every retained candidate is admissible.*

Proof. Modulo $d = j(a+1) + 1$ we have $ja \equiv -(j+1)$, whence

$$j^4 M = (ja)^3 j(2a+1) \equiv (j+1)^3(j+2) = K_j \pmod{d}.$$

Since $\gcd(d, j) = 1$, we have $d \mid M$ if and only if $d \mid K_j$. Thus e is integral. Since $M \equiv d \equiv 1 \pmod{h}$, also $e \equiv 1 \pmod{h}$, so v is integral. The identity

$$M = 1 + h(3a-2) + h^2 C = (1+hj)(1+hv)$$

gives $j + v + hjv = 3a - 2 + hC$. Defining $b = C - jv$, we obtain $d + e = h^2 b + a(3a+1)$ and $e - d = h(v-j)$. The ordering condition gives $v - j \geq 0$; Proposition 6.1 now proves admissibility and $D = (v-j)^2$. The reverse direction follows by applying these identities to any admissible index- j pair. □

This is a complete finite problem for each fixed j , without a truncation in a . Section 7 combines the first three such problems with a uniform argument for all later indices. For $j = 2$, $K_2 = 108$ and $d = 2a + 3$ is an odd divisor of 108 with $d \geq 7$. The only candidates are $d = 9, 27$, giving $a = 3, 12$. The first has $e = 21, v = 5, b = 0$ and is excluded. The second has $e = 1600, v = 123, b = 7$ and $\sqrt{D} = 121$. Thus $(12, 7)$ is the only positive- b case with $j = 2$.

Its symmetric cubic is $f_{12,7}(x) = x^3 - 751x^2 + 180348x - 13829760$. Its values at all residues modulo 13 are 4, 7, 9, 3, 8, 4, 10, 6, 11, 5, 7, 10, 7. As a monic cubic with no root modulo 13, it is irreducible over $\mathbb{Q}$. Its real nonzero roots lift to noninteger graph eigenvalues, settling the entire index-2 case.

Theorem 6.7 (Third linear cutoff). *For every $a \geq 2$, put $T(a) = 2(a-4)(a-2)/(3a+4)$. If $b \geq 1$ and $b > T(a)$, then $G_{p^a q^a r^b}$ is not Laplacian integral. Every unresolved square-discriminant pair has $j \geq 3$, $d \geq 3a+4$, and $b \leq \lfloor T(a) \rfloor$.*

Proof. Nonsquare D gives irrational antisymmetric eigenvalues. For square D , the cases $j = 0, 1, 2$ are all settled above. Every other pair has $d \geq Q = 3a+4$, and $e \geq d \geq Q$ gives

$$b \leq \frac{M/Q + Q - a(3a+1)}{(a+1)^2} = \frac{2(a-4)(a-2)}{3a+4} = T(a).$$

This proves the claimed implication. □

For $a \geq 3$ the new bound is strictly below $S(a)$, since

$$S(a) - T(a) = \frac{2a^3 - a^2 - 5a - 12}{(2a+3)(3a+4)} > 0.$$

With $z = a - 3 \geq 0$, the numerator is $2z^3 + 17z^2 + 43z + 18$. The exact expression $T(a) = 2a/3 - 44/9 + 320/(9(3a+4))$ shows its leading term.

7 Completion of the repeated-exponent family

The preceding factor-index reduction makes the repeated-exponent problem finite at its first few levels. A uniform cubic sign interval handles every later level, completing the family without a bound on either exponent.

Theorem 7.1. *For distinct primes p, q, r and all integers $a, b \geq 1$, the graph $G_{p^a q^a r^b}$ is not Laplacian integral. The same conclusion holds for every permutation of the exponent vector (a, a, b) .*

The arithmetic reduction and a uniform sign obstruction for the cubic combine to exhaust the family. We first settle one last finite factor level.

Lemma 7.2. *Every square-discriminant pair with index $j = 3$ gives a nonintegral Laplacian spectrum.*

Proof. Proposition 6.6 gives $d \mid K_3 = 320$ and $d = 3a + 4$. For $a \geq 2$, the complete divisor list of this shape is 10, 16, 40, 64, 160. The resulting values are

d	a	e	v	b	Outcome
10	2	4	1	0	$d > e$
16	4	36	7	0	$b = 0$
40	12	1080	83	4	Root-free modulo 5
64	20	5125	244	9	Root-free modulo 7
160	52	92274	1741	30	Root-free modulo 7

The three positive- b cubics are

$$\begin{aligned} f_{12,4}(x) &= x^3 - 568x^2 + 100032x - 5248512, \\ f_{20,9}(x) &= x^3 - 1769x^2 + 992820x - 174744000, \\ f_{52,30}(x) &= x^3 - 13394x^2 + 57771168x - 80229951360. \end{aligned}$$

Their values at all residues of the indicated primes are, respectively, $(3, 3, 3, 4, 2)$, $(3, 2, 4, 1, 6, 4, 1)$ and $(1, 6, 4, 1, 3, 2, 4)$. Each monic cubic is therefore irreducible over $\mathbb{Q}$ and has noninteger real roots. Section 2 supplies the lift to noninteger graph eigenvalues. $\square$

Lemma 7.3 (A uniform consecutive-integer interval). *For $a, b \geq 1$, put*

$$L(a) = \frac{a^3 + 2a^2 + 2a + 1}{a(2a + 1)}.$$

If $b < L(a)$, then $f_{a,b}$ has a root strictly between $2ab - 1$ and $2ab$.

Proof. Substitution in (3) gives

$$\begin{aligned} f_{a,b}(2ab) &= 2a^2b^2 > 0, \\ f_{a,b}(2ab - 1) &= -(a + b + 1)(a^3 + 2a^2 + 2a + 1 - a(2a + 1)b). \end{aligned}$$

The second value is strictly negative when $b < L(a)$. The intermediate value theorem gives the asserted noninteger, nonzero root. $\square$

Proof of Theorem 7.1. For $a = 1$, Corollary 6.2 settles every positive b . For $a \geq 2$, nonsquare D gives irrational antisymmetric eigenvalues. When D is square, index $j = 0$ is Theorem 5.1, and indices $j = 1, 2$ are settled in Section 6. Lemma 7.2 settles $j = 3$.

It remains to consider $j \geq 4$. Then $d \geq Q = 4a + 5$, and $e \geq d \geq Q$ implies

$$\frac{M}{Q} + Q - (d + e) = (d - Q) \left(\frac{e}{Q} - 1 \right) \geq 0.$$

Since $d + e = (a + 1)^2b + a(3a + 1)$, we obtain

$$b \leq B_4(a) = \frac{M/Q + Q - a(3a + 1)}{(a + 1)^2} = \frac{(a - 5)(2a - 5)}{4a + 5}.$$

For every $a \geq 2$,

$$L(a) - B_4(a) = \frac{41a^3 - 17a^2 - 11a + 5}{a(2a + 1)(4a + 5)} > 0.$$

Indeed, with $z = a - 2 \geq 0$, the numerator becomes $41z^3 + 229z^2 + 413z + 243$. Thus $b \leq B_4(a) < L(a)$, and Lemma 7.3 gives a noninteger cubic root, which lifts across the universal vertices. These cases exhaust all positive a, b . Permuting the prime factors preserves the graph's isomorphism type. $\square$

The preceding cutoffs and fixed-exponent theorems are intermediate consequences of the arithmetic reduction. The uniform interval now settles the infinite range left by those cutoffs, without a parameter scan.

8 A unit exponent with two arbitrary exponents

The complement quotient replaces coordinate symmetry when the other two exponents are arbitrary. A root between zero and one then settles the entire unit-exponent family after the small exceptions are identified.

Theorem 8.1. *For distinct primes p, q, r and every $b, c \geq 1$, the graph $G_{pq^b r^c}$ is not Laplacian integral. The same holds for permutations of the exponent vector $(1, b, c)$.*

The equal-exponent decomposition is replaced by a complement identity on all six support classes. For arbitrary $a, b, c \geq 1$, remove the $u = abc - 1$ universal vertices and denote the remaining graph by H . In the order $1, 2, 3, 12, 13, 23$, its class-size vector is $w = (a, b, c, ab, ac, bc)^T$ and its vertex count is $N = a + b + c + ab + ac + bc$. Disjoint supports are adjacent in the complement $\overline{H}$, whose cell-constant Laplacian quotient is

$$C = \begin{pmatrix} b+c+bc & -b & -c & 0 & 0 & -bc \\ -a & a+c+ac & -c & 0 & -ac & 0 \\ -a & -b & a+b+ab & -ab & 0 & 0 \\ 0 & 0 & -c & c & 0 & 0 \\ 0 & -b & 0 & 0 & b & 0 \\ -a & 0 & 0 & 0 & 0 & a \end{pmatrix}.$$

Multiplication on the left by $\text{diag}(w)$ makes C symmetric. Moreover $C\mathbf{1} = 0$ and $w^T C = 0$. If B is the quotient of H , the complement relation gives

$$B + C = NI - \mathbf{1}w^T. \quad (8)$$

Thus an eigenvector $Cv = \mu v$ with $\mu \neq 0$ satisfies $w^T v = 0$ and $Bv = (N - \mu)v$. When $N - \mu \neq 0$, Lemma 2.1 lifts it to an eigenvalue $N + u - \mu$ of G_n .

Proof of Theorem 8.1. Write $\det(xI - C) = xh_{a,b,c}(x)$. Direct determinant expansion yields

$$\begin{aligned} h_{a,b,c}(0) &= -abc(a + b + c)N, \\ h_{a,b,c}(a) &= -a^2b^2c^2(a^2 + 2a - b - c). \end{aligned}$$

At $a = 1$, the first value is strictly negative and

$$h_{1,b,c}(1) = b^2c^2(b + c - 3).$$

When $b + c > 3$, a real root μ lies strictly between 0 and 1. Equation (8) and the universal-vertex lift produce a Laplacian eigenvalue $V - \mu \in (V - 1, V)$, where $V = N + u$ is the integer vertex count. This eigenvalue is noninteger. When $b + c \leq 3$, the positive pairs are $(1, 1), (1, 2), (2, 1)$; Theorem 4.1 settles the corresponding permutations of $(1, 1, k)$. Permuting the prime factors preserves the graph's isomorphism type. $\square$

The general six-support reduction also gives an exact diagnostic for fully distinct triples. For $1 \leq a < b < c \leq 7$, all thirty-five quintics have a nonlinear irreducible factor over $\mathbb{Q}$. The quintics at $(1, 2, 7)$ and $(2, 3, 5)$ have factor degrees $(1, 4)$, and that at $(2, 4, 6)$ has factor degrees $(2, 3)$; irreducibility of the entire quintic is therefore not a uniform obstruction. Theorem 8.1 covers every triple beginning with 1 without a range restriction; the thirty-five-case calculation itself remains finite evidence.

9 A minimum exponent of two

Specializing the complement quintic to a minimum exponent of two reduces its endpoint signs to positive polynomial expansions. One remaining pair is handled by a rational interval, completing this minimum uniformly.

Theorem 9.1. *For distinct primes p, q, r and all $b, c \geq 1$, the graph $G_{p^2q^brc}$ is not Laplacian integral. Consequently every three-prime exponent vector with minimum entry at most two gives a nonintegral graph.*

Proof. Use the complement quotient and quintic from Section 8. For $a = 2$, $h_{2,b,c}(0) < 0$. Putting $s = b + c$ and $t = bc$ gives

$$h_{2,b,c}(1) = (s - 13)t^2 - (2s + 6)t + 3s^3 + s^2 - 3s - 1.$$

By symmetry order $b \leq c$. A unit exponent is covered by Theorem 8.1; if $b = 2$ or $b = c$, use Theorem 7.1. It remains to consider $3 \leq b < c$.

When $b = 3$ and $c = 5 + v$, $v \geq 0$, the exact expansion is

$$h_{2,3,5+v}(1) = 12v^3 + 112v^2 + 268v + 120 > 0.$$

When $b \geq 4$ and $c > b$, write $b = 4 + u$ and $c = 5 + u + v$, with $u, v \geq 0$. Then

$$\begin{aligned} h_{2,4+u,5+u+v}(1) &= (u^2 + 8u + 19)v^3 \\ &\quad + (4u^3 + 38u^2 + 128u + 170)v^2 \\ &\quad + (5u^4 + 62u^3 + 271u^2 + 502u + 368)v \\ &\quad + 2u^5 + 32u^4 + 190u^3 + 504u^2 + 552u + 160 > 0. \end{aligned}$$

All coefficients are positive. In either region the opposite signs at 0 and 1 give a root $\mu \in (0, 1)$ and, by (8) and the universal-vertex lift, a noninteger graph eigenvalue in $(V - 1, V)$.

The only remaining pair is $(b, c) = (3, 4)$. Here

$$h_{2,3,4}(x) = x^5 - 53x^4 + 953x^3 - 6523x^2 + 13134x - 7560,$$

and $h_{2,3,4}(1) = -48$ while $h_{2,3,4}(3/2) = 13437/32 > 0$. A root lies in $(1, 3/2)$. Since $V = 58$, its lifted graph eigenvalue lies in $(113/2, 57)$, an interval containing no integer. This exhausts all positive b, c . Combining with Theorem 8.1 proves the minimum-entry assertion. $\square$

10 A uniform cutoff and a minimum exponent of three

The first endpoint polynomial gives a cutoff depending only on the minimum exponent. At minimum three, the finite region below that cutoff is exhausted by an exact certificate, so the whole minimum-three family follows.

Proposition 10.1. *For distinct primes p, q, r and positive exponents ordered as $2 \leq a \leq b \leq c$, the graph $G_{p^a q^b r^c}$ is not Laplacian integral whenever $c \geq 4a^2 - 2a$.*

Proof. Use the complement quintic $h = h_{a,b,c}$ of Section 8. Its first endpoint is the cubic

$$h(1) = Lc^3 + Qc^2 + Rc + S, \tag{9}$$

where

$$\begin{aligned} L &= a^2 + b^2 - 1, \\ Q &= a^3 - 4a^2b^2 + 2a^2b - a^2 + 2ab^2 + ab - 3a + b^3 - b^2 - 3b + 3, \\ R &= (a-1)(b-1)(2ab+3a+3b-3), \\ S &= (a-1)(b-1)(a+b-1)(ab+a+b-1). \end{aligned}$$

Here $L, R, S > 0$. In fact the sharper condition $Lc + Q \geq 0$ already gives $h(1) > 0$. Put $T(a) = 4a^2 - 2a$. Then

$$Q + T(a)L = b^2(b-1) + (2a^2 + a - 3)b + K(a),$$

where $K(a) = 4a^4 - a^3 - 5a^2 - a + 3$. For $a = 2 + u$, $u \geq 0$,

$$K(2+u) = 4u^4 + 31u^3 + 85u^2 + 95u + 37 > 0.$$

Consequently $Lc + Q > 0$ for $c \geq T(a)$. Since $h(0) < 0$, a root lies in $(0, 1)$, producing a noninteger graph eigenvalue in $(V-1, V)$ by (8) and Lemma 2.1. $\square$

Theorem 10.2. *For distinct primes p, q, r and all $b, c \geq 1$, the graph $G_{p^3q^brc}$ is not Laplacian integral. Consequently every three-prime exponent vector with minimum entry at most three gives a nonintegral graph.*

Proof. Theorems 8.1, 9.1 and 7.1 leave $4 \leq b < c$. Proposition 10.1 settles $c \geq 30$. At $a = 3$, equation (9) is

$$\begin{aligned} h_{3,b,c}(1) &= (b^2 + 8)c^3 + (b-1)(b^2 - 30b - 12)c^2 \\ &\quad + 6(b-1)(3b+2)c + 4(b-1)(b+2)(2b+1). \end{aligned}$$

The remaining domain $4 \leq b < c < 30$ has exactly $\binom{26}{2} = 325$ pairs. Exhaustive integer evaluation of this cubic gives 257 positive and 68 negative values, with no zero. Table 1 lists every negative range; all other pairs in this finite domain have positive values. The complete polynomial and separate Horner sign certificates are in `results/distinct-tail.json`; their construction is described in Section A. Thus a positive value gives a root in $(0, 1)$.

b	4	5	6	7	8	9	10	11	12	13
$c_{\min}$	5	6	7	8	9	10	11	12	13	14
$c_{\max}$	13	15	16	17	17	16	16	15	14	14

Table 1: Every negative $h_{3,b,c}(1)$ in $4 \leq b < c < 30$; each column includes all integers between its two endpoints.

For the negative values, the second endpoint is positive except at $(b, c) = (4, 5)$. Indeed, for $v \geq 0$,

$$h_{3,4,6+v}(2) = 104v^3 + 1238v^2 + 4018v + 2344 > 0.$$

For $b \geq 5$, write $b = 5 + u$, $c = 6 + u + v$, with $u, v \geq 0$. Direct expansion gives

$$\begin{aligned} h_{3,5+u,6+u+v}(2) &= (5u^2 + 54u + 153)v^3 \\ &\quad + (20u^3 + 262u^2 + 1179u + 1863)v^2 \\ &\quad + (25u^4 + 426u^3 + 2642u^2 + 7011u + 6624)v \\ &\quad + 10u^5 + 218u^4 + 1828u^3 + 7200u^2 + 12600u + 6480 > 0. \end{aligned}$$

These regions exhaust $4 \leq b < c$ except $(4, 5)$. Whenever $h(1) < 0 < h(2)$, a root lies in $(1, 2)$, giving a noninteger graph eigenvalue in $(V - 2, V - 1)$.

Finally

$$h_{3,4,5}(x) = x^5 - 83x^4 + 2327x^3 - 24133x^2 + 60576x - 42480.$$

Here $h(1) = -3792$ and $h(3/2) = 48825/32 > 0$. The root in $(1, 3/2)$ lifts to a graph eigenvalue in $(233/2, 117)$, since $V = 118$. Every interval used contains no integer. The finite nonvanishing certificate and the written infinite-range arguments exhaust all b, c ; Theorems 8.1 and 9.1 give the final assertion. $\square$

11 An inertia criterion and four families with bounded exponent gaps

An endpoint sign change need not detect two roots in the same interval. For example, the complement quintic at equal exponents factors as

$$h_{a,a,a}(x) = (x - a^2 - a)(x^2 - (a^2 + 4a)x + 3a^2)^2.$$

We instead count low eigenvalues by a symmetric Schur complement.

Proposition 11.1. *Order the exponents as $a \leq b \leq c$, and let m be an integer with $2 \leq m < a$. If*

$$(c - m)(a + b + c - m) < mab, \tag{10}$$

then the six-support complement quotient has exactly two nonzero eigenvalues in $(0, m)$. If, in addition,

$$(a - m + 1)(a + b + c - m + 1) > (m - 1)bc, \tag{11}$$

both lie in $(m - 1, m)$, so $G_{p^a q^b r^c}$ is not Laplacian integral.

Proof. Put $s = a + b + c$ and $P = abc$. Symmetrize C by the square roots of its cell sizes and reorder the classes as 1, 2, 3, 23, 13, 12. The resulting real symmetric matrix is

$$C = \begin{pmatrix} A & -\sqrt{P}I \\ -\sqrt{P}I & D \end{pmatrix}, \quad D = \text{diag}(a, b, c),$$

where A has diagonal $(b + c + bc, a + c + ac, a + b + ab)$ and off-diagonal entries $A_{ij} = -\sqrt{a_i a_j}$, with $(a_1, a_2, a_3) = (a, b, c)$. For $0 < x < a$, the block $D - xI$ is positive definite, and Sylvester's law of inertia gives

$$\text{inertia}(C - xI) = \text{inertia}(D - xI) + \text{inertia}(K(x)),$$

where

$$K(x) = \text{diag}(k_a(x), k_b(x), k_c(x)) - vv^T, \quad v = (\sqrt{a}, \sqrt{b}, \sqrt{c})^T, \quad k_t(x) = s - x - \frac{xP}{t(t - x)}.$$

For fixed $x > 0$, the function $k_t(x)$ increases with $t > x$, since

$$\frac{\partial k_t(x)}{\partial t} = \frac{xP(2t - x)}{t^2(t - x)^2} > 0.$$

Condition (10) is exactly $k_c(m) < 0$. Thus all three diagonal values are negative, so $K(m)$ is negative definite. The matrix $C - mI$ has three negative and three positive eigenvalues. The complement

support graph is a triangle with one pendant cell at each vertex; it is connected, so C has a simple zero eigenvalue and all other eigenvalues are positive. Exactly two nonzero eigenvalues therefore lie in $(0, m)$.

For $x = m - 1$, condition (11) says $k_a(x) > 0$, so all three $k_t(x)$ are positive. Congruence by the inverse square roots of these values turns $K(x)$ into $I - zz^T$, where

$$\|z\|^2 = \frac{a}{k_a(x)} + \frac{b}{k_b(x)} + \frac{c}{k_c(x)} > 1,$$

because each $k_t(x) < s$. Consequently $K(x)$ has one negative and two positive eigenvalues, and the only eigenvalue of C below x is zero. Both low positive eigenvalues lie in $(m - 1, m)$. They are less than $a < N$, so the complement identity and universal-vertex lift apply. Their graph eigenvalues lie in $(V - m, V - m + 1)$, which contains no integer. $\square$

Theorem 11.2. *For every $a \geq 1$, each of the exponent triples*

$$(a, a + 1, a + 2), \quad (a, a + 1, a + 3), \quad (a, a + 2, a + 3), \quad (a, a + 3, a + 4)$$

gives a nonintegral ideal intersection graph on three distinct primes.

Proof. For $a \leq 3$, use Theorem 10.2. Assume $a \geq 4$ and write the triple as $(a, a + d, a + e)$, where $(d, e) = (1, 2), (1, 3), (2, 3), (3, 4)$. The left side minus the right side of (10), at $m = 3$, is respectively

$$-6a, \quad -2a, \quad -4a, \quad 4 - 2a.$$

These values are negative. Proposition 11.1 therefore places exactly two nonzero complement eigenvalues in $(0, 3)$. It remains to exclude the integers 1 and 2.

The first endpoints have the identities $h_{a,a+d,a+e}(1) = -4M_{d,e}(a)P_{d,e}(a)$ from Table 2. Every multiplier is positive for $a \geq 4$, and every coefficient of $P_{d,e}(4 + u)$ is positive. Thus $h(1) < 0$ in each family.

(d, e)	$M_{d,e}(a)$	Descending coefficients of $P_{d,e}(4 + u)$
(1, 2)	$a(a + 1)$	(1, 18, 116, 314, 294)
(1, 3)	$a(a + 1)(a + 3)$	(1, 13, 51, 57)
(2, 3)	$a + 2$	(1, 25, 242, 1126, 2484, 2022)
(3, 4)	$a + 3$	(1, 28, 302, 1550, 3699, 3126)

Table 2: Exact first-endpoint identities and positive shifted factors.

The second endpoints factor as

$$\begin{aligned} h_{a,a+1,a+2}(2) &= -aF_{1,2}(a), \\ h_{a,a+1,a+3}(2) &= -(a - 2)(a + 1)F_{1,3}(a), \\ h_{a,a+2,a+3}(2) &= -aF_{2,3}(a), \\ h_{a,a+3,a+4}(2) &= -F_{3,4}(a), \end{aligned}$$

where

$$\begin{aligned} F_{1,2}(a) &= a^5 - 9a^4 - 14a^3 + 93a^2 + 67a + 6, \\ F_{1,3}(a) &= a^4 - 6a^3 - 42a^2 - 39a - 10, \\ F_{2,3}(a) &= a^5 - 5a^4 - 48a^3 - 59a^2 - 39a - 54, \\ F_{3,4}(a) &= a^6 - a^5 - 74a^4 - 375a^3 - 967a^2 - 1220a - 340. \end{aligned}$$

Table 3 evaluates each factor at every residue of one prime. No value is zero, so no factor has an integer root. The linear multipliers are nonzero for $a \geq 4$, hence $h(2) \neq 0$. Both positive eigenvalues in $(0, 3)$ are therefore noninteger, and their graph lifts are noninteger as well.

Factor	Prime	Values at residues $0, \dots, p-1$
$F_{1,2}$	7	(6, 4, 1, 5, 6, 5, 1)
$F_{1,3}$	11	(1, 3, 9, 8, 2, 6, 4, 4, 5, 8, 5)
$F_{2,3}$	7	(2, 6, 5, 3, 5, 4, 3)
$F_{3,4}$	13	(11, 1, 3, 4, 3, 1, 9, 9, 11, 11, 6, 1, 8)

Table 3: Complete root-free modular certificates for the second endpoints.

□

Corollary 11.3. *Every positive three-prime exponent triple with maximum minus minimum at most three gives a nonintegral ideal intersection graph.*

Proof. Repeated entries are covered by Theorem 7.1. After sorting distinct entries, the only possibilities have gaps $(d, e) = (1, 2), (1, 3), (2, 3)$, all covered by Theorem 11.2. □

11.1 The complete finite region for minimum exponents four through seven

The uniform cutoff makes the requested diagnostic range complete for each of the minimum exponents 4, 5, 6, 7. Here the relevant quintic is $g_B(x) = \det(xI - B)/x$ for H , whereas the preceding endpoint arguments use the complement quintic $h_C(x) = -g_B(N - x)$.

Corollary 11.4. *Every positive three-prime exponent triple with minimum entry at most seven gives a nonintegral ideal intersection graph.*

Proof. Theorem 10.2 handles minimum entries at most three. For a minimum entry $a \in \{4, 5, 6, 7\}$, repeated entries are covered by Theorem 7.1, and Proposition 10.1 settles $c \geq 4a^2 - 2a$. The remaining ordered triples lie in the exact finite domain $a < b < c < 4a^2 - 2a$.

For all 27 562 triples in this domain, the integer discriminant $\Delta_{a,b,c}$ of g_B is positive and strictly between two consecutive integer squares. Table 4 gives the complete counts. The certificate `results/open-region-discriminants.csv` records every triple, discriminant and integer q satisfying $q^2 < \Delta_{a,b,c} < (q+1)^2$. Each discriminant is independently recomputed as a 9×9 integer Sylvester determinant by `scripts/check_open_region_certificate.py`, using fraction-free elimination and exact divisions. The certificate exhausts the domain without omitted or duplicate triples.

If all roots of the monic quintic g_B were integers, its discriminant, the product of the squared pairwise root differences, would be an integer square. Thus every certified quintic has a noninteger root. The quotient is similar to a real symmetric Laplacian, so the root is real and nonzero; Lemma 2.1 gives a noninteger graph eigenvalue. This complete finite certificate and the written cutoff exhaust all triples with minimum entry at most seven. □

The signs of $(g_B(1), g_B(2), g_B(3))$ are $(-, -, -)$ throughout this finite range. The complement signs instead have three patterns, $(-, +, -), (-, +, +), (+, +, +)$, recorded separately in the diagnostic. Different endpoint patterns do not rule out a uniform proof, and equal signs can miss two roots in one interval. Proposition 11.1 addresses that issue without extrapolating from these finite counts.

Minimum a	4	5	6	7	Total
Exclusive cutoff	56	90	132	182	—
Complete pairs	1275	3486	7750	15051	27562
Nonsquare discriminants	1275	3486	7750	15051	27562

Table 4: Exhaustive discriminant certificates in the region derived from the uniform cutoff.

12 A uniform low eigenvalue and a balanced region

The Schur complement of Proposition 11.1 also counts roots when its diagonal has mixed signs. A comparison at three gives a low eigenvalue for every triple with minimum exponent at least four.

Lemma 12.1. *For $0 < x < a$, suppose all three $k_t(x)$ are nonzero. Let q be the number of negative diagonal values and put*

$$F(x) = 1 - \frac{a}{k_a(x)} - \frac{b}{k_b(x)} - \frac{c}{k_c(x)}.$$

Then $n_-(K(x)) = q + \mathbf{1}_{\{F(x) < 0\}}$ and nullity $K(x) = \mathbf{1}_{\{F(x) = 0\}}$. The number of positive complement quotient eigenvalues strictly below x is $n_-(K(x)) - 1$.

Proof. The bordered matrix

$$\begin{pmatrix} \text{diag}(k_t(x)) & v \\ v^T & 1 \end{pmatrix}$$

is congruent both to $K(x) \oplus [1]$ and to $\text{diag}(k_t(x)) \oplus [F(x)]$. Sylvester's law of inertia gives the formulas. The block $D - xI$ is positive definite, and C has a simple zero eigenvalue; subtracting this eigenvalue gives the stated count. $\square$

A direct comparison at three avoids the reciprocal formula at zero diagonal entries and supplies a uniform low positive root.

Theorem 12.2. *For every ordered triple $4 \leq a \leq b \leq c$, the six-support complement quotient has a nonzero eigenvalue in $(0, 3)$.*

Proof. Use $s = a + b + c$, $P = abc$ and $v = (\sqrt{a}, \sqrt{b}, \sqrt{c})^T$ from Proposition 11.1, and define

$$L = \text{diag} \left(s - \frac{3P}{a^2}, s - \frac{3P}{b^2}, s - \frac{3P}{c^2} \right) - vv^T.$$

Expansion of this diagonal-minus-rank-one determinant gives

$$\begin{aligned} \det L &= 6(c(a-b)^2 + b(a-c)^2 + a(b-c)^2), \\ v^T L v &= -3P \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) < 0. \end{aligned}$$

If the exponents are not all equal, the determinant is positive. The negative Rayleigh value supplies a negative eigenvalue, so the positive determinant forces exactly two negative eigenvalues. If all exponents are equal, $L = -vv^T$, with one negative and two zero eigenvalues. In both cases there is a two-dimensional subspace on which L is nonpositive.

Since $a > 3$, the identity

$$K(3) = L - \text{diag}_{t \in \{a, b, c\}} \left(3 + \frac{9P}{t^2(t-3)} \right)$$

subtracts a positive definite matrix. Thus $K(3)$ is strictly negative on that subspace and has at least two negative eigenvalues. As $D - 3I$ is positive definite, $C - 3I$ also has at least two negative eigenvalues. The connected complement quotient has a simple zero eigenvalue and all other eigenvalues are positive. At least one of them therefore lies strictly between zero and three. This argument also covers zero values among the $k_t(3)$. $\square$

Corollary 12.3. *For $4 \leq a \leq b \leq c$, if*

$$(a - 2)(a + b + c - 2) > 2bc, \quad (12)$$

then C has an eigenvalue in $(2, 3)$, and $G_{p^a q^b r^c}$ is not Laplacian integral. Nonzero values of both $h_C(1)$ and $h_C(2)$ also imply nonintegrality. Hence $h_C(1)h_C(2) = 0$ is necessary for an integral graph.

Proof. Condition (12) is exactly $k_a(2) > 0$, so all three diagonal values are positive. As in Proposition 11.1, congruence turns $K(2)$ into $I - zz^T$, with $\|z\|^2 = \sum_t t/k_t(2) > 1$, since every $k_t(2) < s$. It has one negative and two positive eigenvalues and no zero eigenvalue. Consequently the only quotient eigenvalue at or below two is zero. The positive root below three from Theorem 12.2 lies in $(2, 3)$ and lifts to $(V - 3, V - 2)$. For the endpoint assertion, a positive root in $(0, 3)$ can be an integer only if it equals one or two, giving the necessary condition. $\square$

Corollary 12.4. *Let a be the minimum exponent and ρ the maximum minus minimum. Every positive three-prime exponent triple with $a \geq 3\rho + 8$ gives a nonintegral ideal intersection graph.*

Proof. Write the ordered triple as $(a, a + d, a + \rho)$ with $0 \leq d \leq \rho$. The left side minus the right side of (12) is

$$a^2 - 2a\rho - 8a - 2\rho^2 - 4\rho + 4 + (\rho - d)(a + 2\rho + 2).$$

The last term is nonnegative. At $a = 3\rho + 8 + u$, $u \geq 0$, the first part is $(\rho + 2)^2 + (4\rho + 8)u + u^2 > 0$. Corollary 12.3 applies. $\square$

For example, $(20, 21, 24)$, $(38, 43, 48)$ and $(308, 350, 408)$ satisfy the span criterion and lie outside the four fixed-gap families of Theorem 11.2. The argument allows an unbounded span and uses no exponent scan or modular endpoint certificate. The necessary endpoint-zero condition is not sufficient for integrality: those triples may still have other noninteger quotient eigenvalues.

13 Arithmetic obstructions from the complement quotient

The uniform low-root bound rules out common exponent divisors greater than two. Reduction of the characteristic polynomial gives further infinite congruence classes, even when another quotient root is an integer.

Theorem 13.1. *Every positive three-prime exponent triple with $\gcd(a, b, c) \geq 3$ gives a nonintegral ideal intersection graph.*

Proof. Put $d = \gcd(a, b, c)$. All entries of the six-support complement quotient C are integer combinations of the exponents and their pairwise products, so C/d is an integer matrix. Every rational eigenvalue of an integer matrix is an integer, by the rational-root theorem for its monic characteristic polynomial. Therefore every integer eigenvalue of C is a multiple of d .

Order $a \leq b \leq c$. If $a = 3$, Theorem 10.2 applies. Otherwise $a \geq 4$, and Theorem 12.2 supplies an eigenvalue $0 < \mu < 3$. No positive multiple of $d \geq 3$ lies in this interval. Hence μ and its graph lift $V - \mu$ are noninteger. $\square$

In particular, if m has exactly three distinct prime factors, then G_{m^d} is not Laplacian integral for every $d \geq 3$.

Theorem 13.2. *A positive three-prime exponent triple gives a nonintegral ideal intersection graph under either of the following conditions:*

1. *all three exponents are odd;*
2. *there is an odd prime ℓ and a nonzero residue t such that $a \equiv b \equiv c \equiv t \pmod{\ell}$ and $t^2 + 8t + 4$ is a quadratic nonresidue modulo ℓ .*

In particular, common residues 1, 3, 4 modulo 5, or 1, 2, 4, 5 modulo 7, imply nonintegrality.

Proof. The monic integer quintic $h_C(x) = \det(xI - C)/x$ would split into linear factors modulo every prime if all its roots were integers. A nonlinear irreducible factor in one reduction therefore supplies a noninteger real complement root and a noninteger graph lift.

If the exponents are odd, reduce the general quotient modulo two:

$$h_C(x) \equiv x(x^2 + x + 1)^2 \pmod{2}.$$

The quadratic has no root in $\mathbb{F}_2$, proving the first assertion. For a common residue t , the equal-exponent identity gives

$$h_C(x) \equiv (x - t^2 - t)(x^2 - (t^2 + 4t)x + 3t^2)^2 \pmod{\ell}.$$

The quadratic discriminant is $t^2(t^2 + 8t + 4)$. For odd ℓ and nonzero t , the assumed nonresidue makes the quadratic irreducible. Modulo five the discriminant factors for $t = 1, 3, 4$ are 3, 2, 2, respectively; the squares are 0, 1, 4. Modulo seven the corresponding values for $t = 1, 2, 4, 5$ are 6, 3, 3, 6, outside the square residues. $\square$

These conditions allow unequal exponents with arbitrarily large ratios or differences. For example, (11, 16, 21), (13, 18, 23) and (14, 19, 24) have gcd one and mixed parity, but meet the modulo-five criterion. The other parity reductions are x^5 for three even exponents and $x^2(x + 1)^3$ for one or two odd exponents. Their complete splitting does not establish integrality.

The arithmetic arguments do not assume that the smallest positive quotient root is noninteger. Scaling also yields an obstruction for even exponents whose binary reductions split completely.

Theorem 13.3. *Every positive three-prime exponent triple with $a \equiv b \equiv c \equiv 2 \pmod{4}$ gives a nonintegral ideal intersection graph.*

Proof. Write $(a, b, c) = (4u + 2, 4v + 2, 4w + 2)$, set $Q = C/2$, and put $f(x) = \det(xI - Q)/x$. The matrix Q is integral. If all eigenvalues of C were integers, those of Q would be rational and hence integers by the rational-root theorem.

Modulo two, Q is block lower triangular. Its upper block has zero diagonal and all off-diagonal entries one, and its lower block is I_3 . Consequently

$$\det(xI - Q) \equiv x(x + 1)^5 \pmod{2}, \quad f(x) \equiv (x + 1)^5 \pmod{2}.$$

Thus all five hypothetical integer roots of f are odd. For every odd integer t , their five factors $t - \lambda_i$ would give $32 \mid f(t)$. On the other hand, expansion of the general scaled quotient gives the coefficientwise polynomial congruence

$$f(2y + 1) \equiv 8(u + v + w + y - 1) \pmod{32}.$$

In particular $f(3) - f(1) \equiv 8 \pmod{32}$, contradicting the required divisibility of both values. A noninteger complement eigenvalue and its noninteger graph lift follow. $\square$

Corollary 13.4. *If the three positive exponents have the same 2-adic valuation, their ideal intersection graph is not Laplacian integral.*

Proof. At valuation zero, Theorem 13.2 applies. At valuation one, Theorem 13.3 applies. At valuation at least two, their common divisor is at least four, so Theorem 13.1 applies. $\square$

For example, $(10, 14, 18)$, $(14, 22, 34)$ and $(18, 26, 42)$ have gcd two and lie below the linear even-tail cutoff of the following section. The new obstruction applies regardless of the minimum exponent or ratios. The necessary endpoint-zero condition of Corollary 12.3 remains insufficient: the already settled boundary triple $(9, 9, 136)$ has $h_C(1) = 0$ and other noninteger quotient roots. The remaining three-prime problem has gcd at most two, at least one even exponent, and lies outside the congruence classes above as well as the earlier proved regions.

13.1 Mixed-parity residue classes

Shifted reductions of the complement quintic organize the mixed-parity restrictions in one statement. The exponent letters in the residue rows identify roles, independently of their increasing order.

Theorem 13.5. *For positive three-prime exponents, the following conclusions hold for every permutation of the indicated residue roles.*

1. *Each of the classes*

$$(1, 3, 2), (1, 7, 2), (3, 7, 2), (3, 5, 6), (3, 7, 6), (5, 7, 6) \pmod{8}$$

is nonintegral. The classes $(3, 3, 2)$ and $(3, 0, 0)$ modulo four are also nonintegral. These exclusions have no size or ratio bound.

2. *For roles $a \equiv 1, b \equiv 3, c \equiv 2 \pmod{4}$, an integral graph requires*

$$a + c \equiv b(b + 2) \pmod{128}, \quad b \equiv 11 \text{ or } 15 \pmod{16}.$$

With $u = (a - 1)/4$ and $t = (a + c - b(b + 2))/128$, it further requires

$$\begin{cases} t + u + v \equiv 1 \pmod{2}, & b = 16v + 11, \\ t + v \equiv 1 \pmod{2}, & b = 16v + 15. \end{cases}$$

3. *For either $(1, 3, 2)$ or $(3, 3, 0)$ modulo four, if $m = \min\{a, b, c\} \geq 4$ and $M = \max\{a, b, c\}$, an integral graph requires*

$$h_C(2) = 0, \quad M < 7m - 16 + \frac{40}{m + 2}.$$

Every violation of a necessary condition proves nonintegrality.

Proof. The case table and complete shifted-polynomial arguments are in Appendix B. The common mechanism is that an integer-root factorization of h_C forces its shifted binary cubics to split into linear factors; their residue and derivative conditions give the first two parts. For the last part, the odd roots would be three modulo four, excluding one. The positive root below three from Theorem 12.2 must then equal two, and Theorem 14.1 gives the maximum bound. $\square$

For one worked example, take the residue roles $(a, b, c) = (19, 27, 10)$, so $(a, b, c) = (4u + 3, 4v + 3, 4w + 2)$ with $(u, v, w) = (4, 6, 2)$. The shifted quintic satisfies

$$h_C(2y + 1)/8 \in \mathbb{Z}[y], \quad h_C(2y + 1)/8 \equiv y^3 \pmod{2}.$$

Integer roots would therefore force all three odd roots to be one modulo four and $16 \mid h'_C(1)$. But $h'_C(1) \equiv 8(u + v + 1) \equiv 8 \pmod{16}$, a contradiction. The same identity proves the whole $(3, 3, 2)$ class in Appendix B. The surviving arithmetic classes in the theorem remain necessary tests, not integral examples.

14 Endpoint surfaces and their reductions

The positive quotient root in $(0, 3)$ reduces possible integer spectra to the two surfaces $h_C(1) = 0$ and $h_C(2) = 0$. Their size, divisor and finite-certificate restrictions are collected together.

Theorem 14.1. *For positive ordered exponents $a \leq b \leq c$, put $R(a) = 7a - 16 + 40/(a + 2)$. Then:*

1. *If $a \geq 4$, then $c \geq R(a)$ gives $h_C(2) > 0$, so $h_C(2) = 0$ requires $c < R(a)$. In this range, all-even triples and permutations of $(1, 3, 2)$ or $(3, 3, 0)$ modulo four are nonintegral when $c \geq R(a)$. For $a \geq 8$, $c \geq 7a - 12$ suffices.*
2. *For $a, b \geq 4$, an endpoint zero requires, respectively,*

$$\begin{aligned} h_C(1) = 0 &\implies c \mid (a - 1)(b - 1)(a + b - 1)(ab + a + b - 1), \\ h_C(2) = 0 &\implies c \mid 2(a - 2)(b - 2)(a + b - 2)(ab + a + b - 2). \end{aligned}$$

3. *Every triple with $b \leq 15$ is nonintegral. Every permutation of $(2, 2, 2)$ or $(1, 2, 2)$ modulo three is also nonintegral.*

Proof. The full endpoint polynomial and positivity proof are in Appendix C. Its constant terms give the divisor conditions. For the remaining ordered pairs $4 \leq a < b \leq 15$, the complete certificate checks all 132 endpoint cubics over 66 pairs: 8527 positive constant-term divisors, including every one of the 7585 divisors above b , give no endpoint zero. Repeated exponents and minima at most three were already settled. Theorem 12.2 then proves the middle-exponent claim. The two modulo-three classes have both endpoint values nonzero, as shown by the reductions in Appendix C. $\square$

Finitely many endpoint-value congruences alone do not eliminate every candidate, even with the listed gcd, valuation and geometric restrictions: Appendix C gives explicit CRT constructions below both maximum bounds and outside the sufficient inertia exclusion. This limits these endpoint tests; another-root obstruction or further arithmetic information can still exclude the candidates. The fully distinct endpoint surfaces remain open.

14.1 The spectrum on the endpoint-one surface

The endpoint-one equation isolates a simple low root and constrains the remaining quartic. The following proof works with positive real weights.

Theorem 14.2. *Let $4 \leq a \leq b \leq c$ be real and suppose $h_C(1) = 0$. Then*

$$b + c \geq 2a^2 - 2a + 1,$$

with equality exactly when $b = a$ and $c = (a - 1)(2a - 1)$. Root one is simple and is the smallest positive quotient root; all four other quotient roots are strictly greater than a . If μ_1 is the smallest of those four roots, then

$$a < \mu_1 < \min\{c, a + b\}.$$

Proof. Put $q = b + c$, $z = bc$ and $K = 4a^2 - 2a + 1$. The generic determinant is

$$\begin{aligned} h_C(1) &= (q - K)z^2 - a(a - 1)(q + 2a - 1)z + P_a(q), \\ P_a(q) &= (a^2 - 1)q^3 + (a^3 - a^2 - 3a + 3)q^2 - 3(a - 1)^2q - (a - 1)^3. \end{aligned}$$

Since $(b - a)(c - a) \geq 0$, we have $z \geq z_0 = a(q - a)$. When $q \leq Q := 2a^2 - 2a + 1$, the coefficient $q - K \leq -2a^2$ is negative, and the difference between the displayed polynomial at z and at z_0 is

$$(z - z_0)[(q - K)(z + z_0) - a(a - 1)(q + 2a - 1)] \leq 0,$$

strictly unless $z = z_0$. Set $t = q - a \geq a$. At z_0 the value factors as

$$[(a - 1)(2a - 1) - t]B(a, t),$$

where $B = a^3 - 2a^2t^2 + a^2t + a^2 + 3at - 3a + t^2 - 2t + 1$. For $a = 4 + m$, $t = a + u$, with $m, u \geq 0$, the complete identity

$$\begin{aligned} -B(4 + m, 4 + m + u) &= 2m^4 + 4m^3u + 30m^3 + 2m^2u^2 + 47m^2u + 163m^2 \\ &\quad + 16mu^2 + 179mu + 381m + 31u^2 + 222u + 323 \end{aligned}$$

proves $B < 0$. Thus $q < Q$ gives $h_C(1) < 0$. At $q = Q$, equality requires $z = z_0$, hence $b = a$ by ordering, and $c = (a - 1)(2a - 1)$. The factorization also verifies this equality case.

Let $S = W^{1/2}CW^{-1/2}$, with $W = \text{diag}(a, b, c, ab, ac, bc)$. It is symmetric positive semidefinite with a one-dimensional zero kernel. Its pair-support principal block is $\text{diag}(c, b, a)$. For ordered eigenvalues $0 = \lambda_1 < \lambda_2 \leq \dots \leq \lambda_6$, interlacing gives $\lambda_4 \geq a$. The sum bound implies $b + c - (a^2 + 2a) \geq (a - 4)^2 + 4(a - 4) + 1 > 0$, so

$$h_C(a) = -a^2b^2c^2(a^2 + 2a - b - c) > 0.$$

No eigenvalue equals a . The positive determinant $\det(aI - S) = ah_C(a)$ forces an even number of eigenvalues below a , counted with multiplicity. There are at least two, zero and one, and at most three by interlacing. There are therefore exactly two: zero and a simple one. All four remaining roots exceed a .

The pair-support block also gives $\lambda_3 \leq c$. The identity $h_C(c) = -a^2b^2c^2(c^2 + 2c - a - b) < 0$ excludes equality. For the other upper bound, the principal block of S on supports 3, 12, 13, 23 is

$$\begin{pmatrix} ab + a + b & -\sqrt{abc} \\ -\sqrt{abc} & c \end{pmatrix} \oplus (b) \oplus (a).$$

The two-dimensional block has polynomial $Q(x) = x^2 - (ab + a + b + c)x + c(a + b)$, with $Q(a + b) = -ab(a + b) < 0$. Its smaller root lies below $a + b$ and its larger root above it. The four-dimensional principal block thus has exactly three eigenvalues below $a + b$. Interlacing gives $\lambda_3 < a + b$, completing the proof. $\square$

Corollary 14.3. *For integer exponents $4 \leq a \leq b \leq c$ on $h_C(1) = 0$, define $F(x) = h_C(x)/(x-1)$ and $s = a+b+c$, $p = ab+ac+bc$, $r = abc$. Then F is a monic integer quartic with $F(0) = rs(p+s) > 0$. An integer spectrum requires a positive divisor d of $rs(p+s)$ with*

$$a < d < \min\{c, a+b\}, \quad F(d) = 0.$$

Proof. The determinant gives $h_C(0) = -rs(p+s)$, so division by $x-1$ retains the stated nonzero constant. Apply the rational-root theorem to the smallest quartic root and Theorem 14.2. $\square$

All four quartic roots exceed a ; the upper window in the corollary applies to their smallest root. For example, $(8, 8, 105)$ has the quartic factor

$$(x - 968)(x^3 - 1138x^2 + 177145x - 1566600),$$

so another root can exceed both c and $a+b$. Testing the divisor window is finite for each fixed triple. A surviving candidate still requires checking the other factors, and the exponent domain is unbounded.

Remark 14.4. The generic residual $[h_C(x) - xh_C(1)]/(x-1)$ is positive on $[0, 4]$ for all real $a, b, c \geq 8$, including off the endpoint-one surface. A separate complete 350-term positive identity after $a = 8+m$, $b = a+t$, $c = b+u$, $x = 4v/(1+v)$ verifies that statement, including $x = 4$. The shorter proof above suffices for the endpoint-one spectral theorem; the generic residual calculation is retained in the verification note.

15 Complete fixed-exponent families and exact diagnostics

These fixed-exponent diagnostics illustrate how the divisor reductions and the symmetric cubic complement one another. Their exact certificates also distinguish a successful cubic test from a complete spectral obstruction.

Theorem 15.1. *For every $a \in \{2, 3, 4\}$, every $b \geq 1$ and distinct primes p, q, r , the graph $G_{p^a q^a r^b}$ is not Laplacian integral.*

Proof. The cutoff values are $R(2) = 0$, $R(3) = 4/5$, and $R(4) = 2$. Theorem 6.4 settles all b for $a = 2, 3$, and all $b > 2$ for $a = 4$. At $(a, b) = (4, 1)$, $D = 145$ is nonsquare. At $(4, 2)$ the symmetric cubic is

$$f_{4,2}(x) = x^3 - 86x^2 + 2304x - 18816.$$

Its values at $0, 1, 2, 3, 4$ modulo 5 are $4, 3, 1, 4, 3$, so it has no root modulo 5. A cubic over a field without a root is irreducible. Since this cubic is monic, reduction modulo 5 proves irreducibility over $\mathbb{Q}$. Its real nonzero roots lift to noninteger graph eigenvalues. $\square$

The second cutoff reaches the next two fixed exponents with only one remaining discriminant check in each case.

Corollary 15.2. *The same all- b conclusion holds for $a = 5, 6$. Thus it holds for every $a \in \{2, 3, 4, 5, 6\}$.*

Proof. The second cutoff gives $S(5) = 14/13$ and $S(6) = 9/5$, leaving only $b = 1$ in each case. The discriminants are 221 and 313, respectively. Since $14^2 < 221 < 15^2$ and $17^2 < 313 < 18^2$, both give irrational antisymmetric eigenvalues. Every larger b is covered by Theorem 6.5. $\square$

Corollary 15.3. *Every positive b also gives a nonintegral spectrum for $a = 7, 8, 9$. Thus every repeated exponent $a \in \{2, 3, 4, 5, 6, 7, 8, 9\}$ is settled.*

Proof. The third cutoff gives $T(7) = 6/5$, $T(8) = 12/7$, $T(9) = 70/31$. Only four pairs remain under these bounds:

$$(a, b) = (7, 1), (8, 1), (9, 1), (9, 2).$$

Their discriminants are 421, 545, 685, 1489, respectively, between the consecutive squares with lower square roots 20, 23, 26, 38. Each therefore gives irrational antisymmetric eigenvalues, and Theorem 6.7 covers every larger b . $\square$

For comparison, the complete divisor certificates before applying the linear cutoff are given in Table 5. The constants factor as $M = 40 = 2^3 \cdot 5$, $189 = 3^3 \cdot 7$, and $576 = 2^6 \cdot 3^2$. Testing every divisor $d \leq \sqrt{M}$ in (5) gives exactly the displayed admissible pairs. The corresponding cubics in Table 6 have no root modulo the listed primes. These finite certificates supply a second proof using the complete exceptional list. Three pairs are in the boundary family; $(4, 2)$ is the additional symmetric-block obstruction.

a	M	(d, e)	b	$\sqrt{D}$
2	40	(1, 40)	3	13
3	189	(1, 189)	10	47
4	576	(6, 96)	2	18
4	576	(1, 576)	21	115

Table 5: Every admissible factor pair for $a = 2, 3, 4$.

(a, b)	$f_{a,b}(x)$	Witness prime
(2, 3)	$x^3 - 47x^2 + 702x - 3312$	5
(3, 10)	$x^3 - 187x^2 + 11220x - 214200$	13
(4, 2)	$x^3 - 86x^2 + 2304x - 18816$	5
(4, 21)	$x^3 - 485x^2 + 75492x - 3721536$	5

Table 6: Root-free modular reductions for the exceptional cubics.

The separate diagnostic range $a = 2, 3, 4$, $1 \leq b \leq 30$ contains 90 cubics, each with a noninteger root certified in an interval between consecutive integers. The reducible cases in that range are $b = 2, 6$ for $a = 2$, $b = 3, 12$ for $a = 3$, and $b = 4, 20$ for $a = 4$. Each has a linear and an irreducible quadratic factor. These observations do not require whole-cubic irreducibility. Indeed, for every $a \geq 1$,

$$\begin{aligned} f_{a,a}(x) &= (x - 2a(a+1))(x^2 - (5a^2 + 2a)x + 6a^4 + 3a^3), \\ f_{a,a(a+1)}(x) &= (x - 2a(a+1)^2)(x^2 - (3a^3 + 4a^2 + 2a)x + 2a^6 + 6a^5 + 7a^4 + 3a^3). \end{aligned}$$

Consequently no prime can make every diagnostic cubic irreducible: the monic factors in the diagonal family retain positive degree modulo every prime.

For the second family the quadratic discriminant is $a^2(a^4 + 4a + 4)$. For $a \geq 3$ the radicand is between a^4 and $(a^2 + 1)^2$, since $2a^2 - 4a - 3 = 2(a - 3)^2 + 8(a - 3) + 3 > 0$. At $a = 2$ it is 28, between 5^2 and 6^2 . Thus the quadratic is irreducible for every $a \geq 2$.

Remark 15.4. At $a = 1, b = 2$ the cubic is $(x - 3)(x - 6)(x - 8)$, although the full graph has irrational antisymmetric eigenvalues $(13 \pm \sqrt{33})/2$. A direct construction of its 10 vertices gives

$$\det(xI - L) = x(x - 10)(x - 9)^3(x - 7)^2(x - 4)(x^2 - 13x + 34).$$

Thus a noninteger cubic root is a sufficient condition for graph nonintegrality; the converse cannot be used without the other blocks.

16 Remaining questions

The squarefree composite classification and every repeated-exponent triple are settled. For three distinct primes, the remaining ordered cases have

$$8 \leq a < b < c < 4a^2 - 2a, \quad b \geq 16, \quad \gcd(a, b, c) \in \{1, 2\},$$

unequal 2-adic valuations and at least one even exponent. They lie outside the bounded-gap families and satisfy $(a - 2)(a + b + c - 2) \leq 2bc$ and $a < 3(c - a) + 8$. The common-residue exclusions, Theorem 13.5 and the modulo-three classes in Theorem 14.1 impose the additional stated restrictions. The following questions remain open.

1. Can the fully distinct endpoint-one surface be excluded, or can another quotient root be shown noninteger on it? Theorem 12.2 reduces possible integer spectra to $h_C(1)h_C(2) = 0$, but vanishing of one endpoint is not an integrality characterization. On endpoint one, Theorem 14.2 places all four residual roots above a and their smallest below $\min\{c, a + b\}$. Their nonzero constant gives the necessary divisor test in Corollary 14.3; whether the quartic splits entirely into integer roots remains open. For fully distinct triples the new sum restriction $b + c > 2a^2 - 2a + 1$ is strict.
2. Can the endpoint-two surface be excluded inside $c < 7a - 16 + 40/(a + 2)$, subject to the applicable arithmetic restrictions? All-even triples and the mixed patterns $(1, 3, 2)$ and $(3, 3, 0)$ must lie on this surface. For a fixed pair (a, b) the endpoint divisor conditions give finitely many candidates; the minimum a remains unbounded.
3. Can the surviving $(1, 3, 2)$ arithmetic classes be resolved uniformly at the polynomial level? Finite closure by further two-adic shifts is not established. The endpoint-congruence limitation in Appendix C concerns finite endpoint-value tests with the explicitly listed filters; it does not exclude another-root or additional arithmetic obstructions.
4. Which nonsquarefree exponent vectors with more than three prime factors give integral graphs? The present arguments do not supply the full general-exponent Q3 characterization.

The distinction between the repeated-exponent blocks remains useful for these questions. The boundary family has an integral antisymmetric block, while $(1, 1, 2)$ has an integral symmetric cubic and a nonintegral full spectrum. The complete repeated-exponent proof uses both blocks.

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A Exact verification and reproducibility

The infinite conclusions use the written proofs and the finite certificates to which those proofs reduce the problem. Computations use integer or rational arithmetic; no floating-point spectrum or Lean formalization is claimed for these Laplacian results. SymPy 1.14.0 and Python 3.10 or later are used, with source hashes in the saved results.

The project repository contains every checker, exact certificate and reproduction command [16]. The build guide gives the commands by filename. The complete per-checker catalogue is retained in `paper/sections/verification-details.tex`; the optional detailed build includes it without changing the proofs or saved certificates. The principal verification scopes are as follows.

A.1 Graph embeddings and repeated exponents

Direct ideal adjacency validates $LU = UB$, independently of the support construction, on 14 squarefree and $(1, 1, k)$ cases with 44 294 vertex pairs. The full 34-vertex polynomial at $(2, 2, 3)$ is checked. Three boundary graphs with 34, 174, 548 vertices cover 165 490 pairs and verify both repeated-exponent embeddings and the quotient lifting. The 548-vertex full characteristic polynomial is not computed.

The 90 cubic diagnostics have exact rational intervals and independent Sturm/sign checks. The all- b proofs use the exhaustive divisor reduction, not extrapolation from those diagnostics. All divisors of 24, 108, 320 for smaller-factor indices 1, 2, 3 are exhausted; eight positive- b cubics have modular certificates checked by separate integer Horner arithmetic. Higher indices use the uniform sign proof. The separate ten-vertex graph for $(1, 1, 2)$ confirms why the symmetric cubic alone is not an integrality characterization.

A.2 Small minima and complete finite domains

The general six-support quintic agrees with the reflected source quotient and weighted symmetry. Thirty-five requested fully distinct triples and direct ideal graphs with 22, 58 vertices check the construction. Endpoint identities and positive shifted coefficients support the written unit, minimum-two and uniform-cutoff proofs.

At minimum three, the proved cutoff leaves precisely 325 pairs $4 \leq b < c < 30$. All integer quintics, endpoint signs and intervals are checked; the first endpoint has 257 positive and 68 negative values, with no zero. This complete certificate is part of the minimum-three proof. At minima four through seven, all 27 562 ordered triples in the derived region are exhausted. A separate fraction-free Bareiss implementation reconstructs every discriminant by a 9×9 integer Sylvester determinant, checking exact divisions, consecutive-square brackets and domain completeness. Together with the written cutoff, this proves Corollary 11.4; the minimum itself is not extrapolated.

A.3 Inertia and endpoint surfaces

Symbolic block, Schur-complement, determinant and positivity identities support the uniform low root and balanced region. Six secular/Sturm fixtures cover the possible nonsingular sign combinations; nine direct quotient/Sturm fixtures include equal exponents and zero-diagonal boundaries. The four bounded-gap families also have complete 38-residue modular certificates. These finite fixtures check formulas; the infinite conclusions use the written comparison and inertia arguments.

For the middle-exponent bound, all 132 cubics over 66 pairs $4 \leq a < b \leq 15$ are independently reconstructed. Integer trial division exhausts 8 527 positive divisors, and separate Horner evaluation checks all 7 585 above b . No upper bound on c is imposed. This is the complete certificate underlying Theorem 14.1. The linear cutoff uses the general endpoint cubic and five positive coefficient expressions. Both known repeated-exponent endpoint zeros are positive controls. The CRT scope note uses six substitution identities, two positive boundary shifts and two integer seed determinants; its all-moduli claim is the written polynomial-congruence argument.

A.4 Arithmetic and shifted-root cases

Scaling, all parity patterns and common residues for primes 3, 5, 7 are checked coefficientwise, with 70 quadratic-residue evaluations and separate integer Horner arithmetic. The all-even scaling contradiction uses the written valuation proof. Complete endpoint tables modulo two and three

include zero residues and zero polynomials; 70 independent integer determinant evaluations check the endpoint table.

For the cases in Appendix B, direct and reflected quintics and all exponent permutations agree. Every shifted divisibility and binary-cubic identity is checked coefficientwise for arbitrary shifts. Chosen full quintics and derivatives are independently reconstructed by integer determinants and principal cofactors, including compatible controls and both known endpoint-zero examples. The finite residue and parity tables support the identities; the infinite conclusions use the written root factorizations, rather than a scan of exponents.

Reza’s higher-modulus script was run unchanged; the independent verifier checks all 28 032 residue triples with 56 064 Bareiss determinants. The coefficientwise period-eight proof explains why the fixed divisor-32 tables add no classes at larger exponent moduli. His derivative script’s 384 representatives are checked by 16 380 principal-minor determinants, including the additional controls. Zero has infinite valuation, and full valuations need not be constant on an exponent residue class. The complete catalogue retains these diagnostic limits, coefficient counts, fixture counts and exact source/certificate mappings.

A.5 Endpoint-one spectral location

Theorem 14.2 and Corollary 14.3 use written proofs without a finite base. The minimum-root checker reconstructs the generic endpoint polynomial, comparison and equality factorization, all twelve positive terms and the determinant identities. The pair-sum checker verifies the four-support block, characteristic factor, discriminant and strict sign identity. The established controls (8, 8, 105), (9, 9, 136) and (20, 20, 741) leave 5, 5, 13 divisors in the smallest-root windows; all 23 integer Horner evaluations are nonzero. Rational factorization agrees, and exact Sturm counts give one quartic root in each window and none on $[0, a]$. These controls already follow from the repeated-exponent theorem. The source-hashed certificates reproduce byte-for-byte; their finite diagnostics supplement the written bounds.

B Mixed-parity case analyses and two-adic lifting

Theorem 13.5 collects the following cases. All residue-role permutations are included. Each row records either a nonintegral class or a necessary condition for an integer spectrum.

Residue roles	Modulus	Conclusion or necessary condition
Six classes listed in the main theorem	8	Nonintegral by the divisor-32 endpoint table below.
(3, 3, 2)	4	Nonintegral: endpoint and derivative requirements are incompatible.
(3, 0, 0)	4	Nonintegral: one is excluded by root distribution and $h_C(2) \equiv 4 \pmod{8}$.
(1, 3, 2)	4	$a + c \equiv b(b + 2) \pmod{128}$, $b \equiv 11, 15 \pmod{16}$ and the stated binary-cubic parity condition.
(1, 3, 2), (3, 3, 0)	4	For minimum $m \geq 4$, $h_C(2) = 0$ and maximum $M < R(m)$.

The complete identities, intermediate conditions, examples and diagnostic boundaries are retained below. The variables identify residue roles rather than size order; no additional refinement or search is introduced.

B.1 Endpoint divisibility and odd-root distribution

Lemma B.1. *Let $f \in \mathbb{Z}[x]$ be monic of degree five with $f(x) \equiv x^2(x+1)^3 \pmod{2}$. If all its roots are integers, then $32 \mid f(1)$ or $32 \mid f(3)$.*

Proof. The binary factorization forces precisely three odd roots and two even roots, counted with multiplicity. At least two odd roots share a residue $r \in \{1, 3\}$ modulo four. In $f(r)$, their two factors are divisible by four and the third odd-root factor is divisible by two. The even-root factors are odd. Therefore $32 \mid f(r)$, including when $f(r) = 0$. $\square$

Proposition B.2. *Let a, b be the two odd exponents and c the even exponent of a positive three-prime exponent triple. Each row in the following table, including exchange of a, b , implies that its ideal intersection graph is nonintegral.*

$a \bmod 8$	$b \bmod 8$	$c \bmod 8$	$h_C(1) \bmod 32$	$h_C(3) \bmod 32$
1	3	2	8	16
1	7	2	24	16
3	7	2	16	8
3	5	6	8	16
3	7	6	16	24
5	7	6	24	16

Proof. For either mixed exponent parity, $h_C(x) \equiv x^2(x+1)^3 \pmod{2}$. Substituting $(a, b, c) = (8u+r_1, 8v+r_2, 8w+r_3)$ in the general quotient quintic gives the last two columns: every nonconstant coefficient in u, v, w is divisible by 32. Both entries are nonzero modulo 32, contradicting Lemma B.1 if all quotient roots were integers. The resulting noninteger real complement eigenvalue has a noninteger graph lift. $\square$

For example, $(10, 17, 19)$ and $(14, 19, 21)$ meet these criteria with gcd one and unequal 2-adic valuations. There is no minimum or ratio bound in the argument. The weaker divisibilities $4 \mid h_C(2t)$ and $8 \mid h_C(2t+1)$ hold identically for both mixed exponent parities; they alone do not supply the contradiction. The distribution of the three odd roots modulo four is the additional constraint.

Lemma B.3. *Let $f \in \mathbb{Z}[x]$ be monic of degree five with $f(x) \equiv x^2(x+1)^3 \pmod{2}$. Suppose $F(y) = f(2y+1)/8$ belongs to $\mathbb{Z}[y]$ and $F(y) \equiv y^3 \pmod{2}$. If all roots of f are integers, then its three odd roots are congruent to one modulo four, $64 \mid f(1)$ and $16 \mid f'(1)$.*

Proof. Write the two even roots as η_1, η_2 and the three odd roots as $2r_1+1, 2r_2+1, 2r_3+1$, with multiplicities. Then

$$F(y) = \prod_{j=1}^2 (2y+1-\eta_j) \prod_{i=1}^3 (y-r_i).$$

The first two factors reduce to one modulo two. Hence $\prod_i (y-r_i) \equiv y^3 \pmod{2}$, which forces every r_i to be even. Each odd-root factor in $f(1)$ is therefore divisible by four, giving $64 \mid f(1)$. Every term in the product-rule expansion of $f'(1)$ retains at least two such factors, giving $16 \mid f'(1)$. The argument includes coincident roots and zero factors. $\square$

Proposition B.4. *Every positive three-prime exponent triple that is a permutation of $(3, 3, 2)$ modulo four has a nonintegral ideal intersection graph.*

Proof. Relabel the odd exponents as $a = 4u + 3$, $b = 4v + 3$ and the even exponent as $c = 4w + 2$, where $u, v, w \geq 0$. Expansion of the general complement quintic gives the coefficientwise identities

$$\begin{aligned} h_C(2y+1)/8 &\in \mathbb{Z}[u, v, w, y], & h_C(2y+1)/8 &\equiv y^3 \pmod{2}, \\ h_C(1) &\equiv 16(u+v) \pmod{32}, & h'_C(1) &\equiv 8(u+v+1) \pmod{16}. \end{aligned}$$

Its binary reduction is $x^2(x+1)^3$. If all five roots were integers, Lemma B.3 would give $64 \mid h_C(1)$, so the endpoint congruence would require $u+v$ even. The derivative divisibility $16 \mid h'_C(1)$ would require $u+v$ odd, a contradiction. Thus there is a noninteger real complement eigenvalue, whose graph lift is noninteger. $\square$

This theorem contains the two rows $(3, 7, 2)$ and $(3, 7, 6)$ of Proposition B.2 and also covers odd exponents sharing the same residue modulo eight. For example, $(10, 19, 27)$ and $(14, 23, 39)$ have odd residues $(3, 3)$ and $(7, 7)$ modulo eight, respectively. Neither theorem requires a low-root interval or an exponent-size bound.

For the fixed necessary divisor 32, increasing the exponent modulus alone gives no further exclusions. On all mixed-parity triples, the values of both $h_C(1)$ and $h_C(3)$ modulo 32 already have period eight in each exponent. This follows coefficientwise by substituting the two parity representatives $(2u+1, 2v, 2w)$ and $(2u+1, 2v+1, 2w)$ and shifting one variable by $4z$; symmetry covers their permutations. Thus the higher-modulus tables are lifts of the six classes in Proposition B.2. The additional exclusions in Proposition B.4 use the distribution of the roots and the derivative, beyond those two endpoint divisibilities.

Proposition B.5. *Let a, b, c be positive exponent residue roles with $a \equiv 1$, $b \equiv 3$ and $c \equiv 2 \pmod{4}$. If their ideal intersection graph is Laplacian integral, then*

$$a + c \equiv b(b+2) \pmod{128}, \quad b \equiv 11 \text{ or } 15 \pmod{16}.$$

Moreover, put $u = (a-1)/4$ and $t = (a+c-b(b+2))/128$. Then

$$\begin{cases} t + u + v \equiv 1 \pmod{2}, & b = 16v + 11, \\ t + v \equiv 1 \pmod{2}, & b = 16v + 15. \end{cases}$$

If $m = \min\{a, b, c\} \geq 4$ and $M = \max\{a, b, c\}$, then also

$$h_C(2) = 0, \quad M < 7m - 16 + \frac{40}{m+2}.$$

Every violation under the stated hypotheses proves nonintegrality; all exponent permutations are included.

Proof. Proposition B.10 in Appendix B forces all three hypothetical integer odd quotient roots to be b modulo eight. Thus $512 \mid h_C(b)$ and $64 \mid h'_C(b)$. The exact identity

$$h_C(b) = a^2 b^2 c^2 (a + c - b(b+2))$$

has prefactor of 2-adic valuation two, giving the sum condition modulo 128, including when $h_C(b) = 0$. The derivative congruence and the binary cubic calculation in Appendix B give the restriction on b and the two parity conditions.

Since $b \equiv 3 \pmod{4}$, one cannot be a quotient root under the integer-spectrum assumption. For $m \geq 4$, the positive root in $(0, 3)$ from Theorem 12.2 must therefore equal two. Proposition C.1 gives the strict linear upper bound. $\square$

The arithmetic conditions imply the earlier sum congruences $a+c \equiv 15 \pmod{16}$ and $a+c+4b \equiv 27 \pmod{32}$. Appendix B gives their staged derivation and the supporting derivative thresholds. The theorem collects these arithmetic conditions together with the endpoint-two restriction. For example, $(29, 27, 754)$ and $(45, 47, 2258)$ have $h_C(b) = 0$ and $64 \mid h'_C(b)$ but fail the final parity test. The compatible triples $(25, 27, 758)$ and $(29, 27, 882)$ instead fail the linear upper bound. Satisfying the arithmetic conditions alone does not establish integrality.

Lemma B.6. *For positive exponent triples that are permutations of $(3, 0, 0)$, $(1, 3, 2)$ or $(3, 3, 0)$ modulo four, an integer complement quotient spectrum would force every odd root to be three modulo four. Consequently one cannot be a quotient root under this assumption.*

Proof. For each displayed residue pattern, write $(a, b, c) = (4u + r_a, 4v + r_b, 4w + r_c)$. Coefficientwise expansion gives

$$\begin{aligned} h_C(x) &\equiv x^2(x+1)^3 \pmod{2}, \\ F(y) = h_C(2y+1)/8 &\in \mathbb{Z}[u, v, w, y], & F(y) &\equiv (y+1)^3 \pmod{2}. \end{aligned}$$

If every root were an integer, there would be three odd roots $2r_i + 1$ and two even roots η_j , counted with multiplicity. In

$$F(y) = \prod_{j=1}^2 (2y+1-\eta_j) \prod_{i=1}^3 (y-r_i),$$

the first two factors reduce to one modulo two. Thus every r_i is odd, giving the asserted residue and exclusion of one. Symmetry of the quintic includes every permutation of the exponents. $\square$

Proposition B.7. *Every positive three-prime exponent triple that is a permutation of $(3, 0, 0)$ modulo four gives a nonintegral ideal intersection graph, without a size or ratio bound.*

Proof. For residue roles $(a, b, c) = (4u+3, 4v, 4w)$, coefficientwise expansion gives $h_C(2) \equiv 4 \pmod{8}$, so two is not a root. If the minimum exponent is at least four, Theorem 12.2 provides a positive quotient root in $(0, 3)$. Under the integer-spectrum assumption it must be one or two. Lemma B.6 excludes one and the endpoint congruence excludes two, a contradiction. The only possible smaller minimum is three, already covered by Theorem 10.2. A noninteger complement root has a noninteger graph lift. $\square$

The triples $(19, 36, 100)$ and $(35, 52, 120)$ illustrate this one-odd-exponent class beyond the minimum-through-seven and middle-through-fifteen certificates. The result uses the uniform low-root theorem together with the shifted root distribution, without enlarging either finite certificate.

The endpoint distributions above give value divisibilities. Derivatives record additional constraints on the same hypothetical integer roots.

B.2 Derivative divisibility

Lemma B.8. *Let $f \in \mathbb{Z}[x]$ be monic of degree five with two even and three odd integer roots. If exactly k of the odd roots are one modulo four, then*

$$2^{3+k} \mid f(1), \quad 2^{\max(2, k+1)} \mid f'(1), \quad 2^{\max(2, k)} \mid f''(1), \quad 2 \mid f'''(1).$$

At three, the same statements hold with k replaced by $3 - k$.

Proof. At one, the odd-root factors supply k factors divisible by four and $3 - k$ divisible by two; the even-root factors are odd. Each term of the j th derivative is $j!$ times a product retaining $5 - j$ root factors. Removing the odd-root factors with the strongest guaranteed divisibility gives the stated bounds, including the factor two from $2!$ and $3!$. At three, the roles of the two odd residue classes are exchanged. Repeated roots and zero evaluations obey the same divisibilities. $\square$

The derivative thresholds are useful once the odd roots share a residue. We now establish that cluster and compare its first two shifted reductions.

B.3 Odd roots modulo four and eight

Proposition B.9. *Let a, b, c be positive exponents with $a \equiv 1$, $b \equiv 3$ and $c \equiv 2 \pmod{4}$. If $a + c \not\equiv 15 \pmod{16}$, their ideal intersection graph is nonintegral. All permutations are included, without a size or ratio bound.*

Proof. Write $(a, b, c) = (4u + 1, 4v + 3, 4w + 2)$. Coefficientwise expansion gives

$$h_C(2y + 1)/8 \in \mathbb{Z}[u, v, w, y], \quad h_C(2y + 1)/8 \equiv (y + 1)^3 \pmod{2}.$$

As in Lemma B.3, integer quotient roots would force all three odd roots to be three modulo four. Consequently $64 \mid h_C(3)$. A further coefficientwise identity is

$$h_C(3)/16 \in \mathbb{Z}[u, v, w], \quad h_C(3)/16 \equiv (2v + 1)(u + w + 1) \pmod{4}.$$

Since $2v + 1$ is invertible modulo four, this forces $u + w + 1 \equiv 0 \pmod{4}$, equivalently $a + c \equiv 15 \pmod{16}$. Failure of that condition gives a noninteger complement root and graph lift. $\square$

For example, $(14, 25, 27)$ and $(10, 29, 31)$ pass the earlier divisor-32 test but have $h_C(3) \equiv 32 \pmod{64}$, contradicting the new requirement. The sum class $a + c \equiv 15 \pmod{16}$ remains compatible with this test. Complete second- and third-derivative checks at one and three give no additional exclusions with the root information modulo four used here. In particular $h_C'''(1) \equiv 2 \pmod{4}$ for every mixed exponent parity. Full valuations at a chosen exponent representative are not generally constant throughout its residue class: $h_C(3) = -2592$ at $(1, 3, 6)$ but $h_C(3) = 0$ at $(9, 3, 6)$, although the exponent triples agree modulo eight.

Proposition B.10. *Let a, b, c be positive exponents with $a \equiv 1$, $b \equiv 3$ and $c \equiv 2 \pmod{4}$. If $a + c + 4b \not\equiv 27 \pmod{32}$, their ideal intersection graph is nonintegral. All permutations are included, without a size or ratio bound.*

Proof. Suppose all five complement quotient roots are integers. Proposition B.9 requires $a + c \equiv 15 \pmod{16}$. Write

$$(a, b, c) = (4u + 1, 4v + 3, 16t - 4u - 2).$$

Coefficientwise expansion gives

$$H(z) = h_C(4z + 3)/64 \in \mathbb{Z}[u, v, t, z], \quad H(z) \equiv z^3 + vz^2 + vz + t + 1 \pmod{2}.$$

The three odd roots are already known to be three modulo four. Writing them as $4r_i + 3$ and the two even roots as η_j gives

$$H(z) = \prod_{j=1}^2 (4z + 3 - \eta_j) \prod_{i=1}^3 (z - r_i).$$

The first two factors reduce to one modulo two, so the displayed cubic must split into linear factors over $\mathbb{F}_2$. For even v , it splits precisely when t is odd: otherwise it is $(z+1)(z^2+z+1)$. For odd v , it splits precisely when t is even: otherwise it is $z(z^2+z+1)$. The quadratic z^2+z+1 is irreducible. Thus $t+v$ must be odd. Since

$$a+c+4b=16(t+v)+11,$$

this forces the asserted congruence. Its failure gives a noninteger complement root and graph lift. $\square$

This excludes half the classes left by Proposition B.9. There are 512 role-ordered classes modulo thirty-two; the earlier condition leaves 128, and the new one leaves 64. Thus 64 classes, or 384 after all permutations, are additionally excluded. For example, the ordered triples (13, 18, 19) and (9, 31, 38) satisfy the old sum condition and $64 \mid h_C(3)$, but their residue-role weighted sums are eleven modulo thirty-two. The compatible binary cubics are z^3 and $(z+1)^3$, so hypothetical integer odd roots would all be b modulo eight, also requiring $512 \mid h_C(b)$. The surviving congruence classes do not assert integral spectra.

The preceding cluster makes evaluation at the odd exponent itself particularly restrictive. Its exact value and derivative lead to the next binary cubic.

B.4 Value, derivative and the third shifted cubic

Proof of Proposition B.5, arithmetic conditions. The proof of Proposition B.10 forces the three hypothetical integer odd complement roots to be b modulo eight. The two remaining roots are even. Thus $512 \mid h_C(b)$ and $64 \mid h'_C(b)$, since each first-derivative term retains at least two factors divisible by eight. The exact quintic identity

$$h_C(b) = a^2b^2c^2(a+c-b(b+2))$$

has prefactor with 2-adic valuation exactly two. It follows that $128 \mid a+c-b(b+2)$, including when the value is zero.

Write $(a, b, c) = (4u+1, 4w+3, b(b+2)+128t-a)$. Coefficientwise expansion gives

$$h'_C(b) \equiv 16(w^2 - w + 2) \pmod{64}.$$

For $w \equiv 0, 1, 2, 3 \pmod{4}$, the derivative residues are respectively 32, 32, 0, 0. Its required divisibility forces $w \equiv 2$ or $3 \pmod{4}$, which is the stated restriction on b .

For the two remaining residues, put $K(z) = h_C(8z+b)/512$. Coefficientwise identities give $K \in \mathbb{Z}[u, v, t, z]$ and

$$K(z) \equiv \begin{cases} z^3 + z^2 + (1+v+u^2)z + t, & b = 16v + 11, \\ z^3 + z^2 + (1+v)z + t, & b = 16v + 15 \end{cases} \pmod{2}.$$

Writing the odd roots as $b+8r_i$ and the even roots as η_j gives

$$K(z) = \prod_{j=1}^2 (8z+b-\eta_j) \prod_{i=1}^3 (z-r_i).$$

The even-root factors reduce to one, so the displayed binary cubic must split completely. For $\lambda, t \in \mathbb{F}_2$, the cubic $z^3 + z^2 + \lambda z + t$ splits precisely when $t = \lambda$: the two possibilities are $z^2(z+1)$ and $(z+1)^3$. Otherwise it is $z(z^2+z+1)$ or the irreducible cubic z^3+z^2+1 . Using $u^2 \equiv u \pmod{2}$ gives the two stated parity conditions. $\square$

C Endpoint polynomial proofs, certificates and covering scope

The two possible low integer roots have different arithmetic constraints. The endpoint two is confined to a linear region in the minimum exponent; this gives a new nonintegral tail for even exponent triples. The shifted odd-root distributions extend the same tail to two mixed-parity patterns below.

Proposition C.1. *Let $4 \leq a \leq b \leq c$, and put*

$$R(a) = \frac{7a^2 - 2a + 8}{a + 2} = 7a - 16 + \frac{40}{a + 2}.$$

If $h_C(2) = 0$, then $c < R(a)$. In fact, $c \geq R(a)$ implies $h_C(2) > 0$.

Proof. Fix a, c and expand the symmetric endpoint polynomial in b :

$$h_C(2) = A(a, c)b^3 + B(a, c)b^2 + D(a, c)b + E(a, c),$$

where

$$\begin{aligned} A(a, c) &= ac(a + c - 1) + 2a(a - 1) + 2c(c - 1) - 4, \\ B(a, c) &= c^2((a + 2)c - (7a^2 - 2a + 8)) \\ &\quad + (a^3 + 2a^2 - 6a - 4)c + 2(a - 2)(a^2 - 2a - 6), \\ D(a, c) &= (a - 2)(c - 2)(a^2c + a^2 + ac^2 + 6ac + 4a + c^2 + 4c - 12), \\ E(a, c) &= 2(a - 2)(c - 2)(a + c - 2)(ac + a + c - 2). \end{aligned}$$

For $a, c \geq 4$, the coefficients A, D, E are strictly positive. Both $a^3 + 2a^2 - 6a - 4$ and $a^2 - 2a - 6$ are positive at $a = 4$ and increasing on $[4, \infty)$. Thus $c \geq R(a)$ also gives $B > 0$, which proves the strict endpoint sign. Conversely, an endpoint zero requires $B < 0$, so $(a + 2)c < 7a^2 - 2a + 8$. $\square$

Proposition C.2. *If $4 \leq a \leq b \leq c$ are all even and $c \geq R(a)$, then $G_{p^a q^b r^c}$ is not Laplacian integral. For minimum exponent at least eight, $c \geq 7a - 12$ suffices.*

Proof. Every entry of C is even, so every integer eigenvalue of C is even by the rational-root theorem applied to $C/2$. In particular, one is not an eigenvalue. Proposition C.1 excludes two, and Theorem 12.2 provides a noninteger root in $(0, 3)$. Equivalently, $h_C(0) < 0 < h_C(2)$ directly yields a root in $(0, 2)$, which cannot equal one. Its graph lift is noninteger. For $a \geq 8$, $40/(a + 2) \leq 4$ gives $R(a) \leq 7a - 12$. $\square$

The triples $(8, 10, 44)$, $(10, 14, 58)$ and $(40, 42, 266)$ meet this criterion with gcd two. Their largest exponents are below the earlier quadratic cutoff and their spans are outside the earlier balanced region. The already settled repeated triple $(10, 10, 12)$ has $h_C(2) = 0$ and obeys the strict bound; an integer endpoint does not establish integrality.

Proposition C.3. *Let $4 \leq m \leq d \leq M$ be the ordered exponents of a triple that is a permutation of $(1, 3, 2)$ or $(3, 3, 0)$ modulo four. If $M \geq R(m)$, the ideal intersection graph is nonintegral. For $m \geq 8$, the simpler bound $M \geq 7m - 12$ suffices.*

Proof. Under the integer-spectrum assumption, Lemma B.6 excludes one as a quotient root. Theorem 12.2 then forces two to be a root. But Proposition C.1 gives $h_C(2) > 0$ for $M \geq R(m)$, a contradiction. The simpler bound follows from $R(m) \leq 7m - 12$ for $m \geq 8$. $\square$

Thus the linear upper bound also restricts these two mixed-parity patterns. For example, the residue-role triples (25, 27, 758) and (29, 27, 882) satisfy the arithmetic conditions of Proposition B.5, but their largest exponents exceed this linear bound. The root-distribution argument therefore settles these compatible controls without a further congruence lift. The triple (19, 27, 140) illustrates the other pattern. The variables m, d, M identify size order, independently of residue roles.

The size bound confines one surface geometrically. The following constant-term conditions give finite candidate sets for either surface when two exponents are fixed.

Proposition C.4. *For fixed $a, b \geq 4$, any positive integer c on an endpoint-zero surface satisfies*

$$\begin{aligned} h_C(1) = 0 &\implies c \mid (a-1)(b-1)(a+b-1)(ab+a+b-1), \\ h_C(2) = 0 &\implies c \mid 2(a-2)(b-2)(a+b-2)(ab+a+b-2). \end{aligned}$$

Proof. Both endpoints are integer cubics in c . Their constant terms are the displayed nonzero products. Reduce each endpoint equation modulo c . $\square$

For a fixed pair this gives finite divisor candidates, which must still satisfy the endpoint equation. Reza's requested diagnostic for middle exponents through fifteen now has a complete independent certificate.

Proposition C.5. *Every positive ordered three-prime exponent triple $a \leq b \leq c$ with $b \leq 15$ gives a nonintegral ideal intersection graph. This conclusion uses the complete finite endpoint certificate below.*

Proof. Repeated exponents are covered by Theorem 7.1, and minimum exponents one, two and three by Theorems 8.1, 9.1 and 10.2, respectively. It remains to consider $4 \leq a < b \leq 15$ and $c > b$.

For each of these 66 pairs, both endpoint polynomials are integer cubics in c with the positive constant terms in Proposition C.4. The complete endpoint certificate records all 132 coefficient lists and exhausts their 8 527 positive divisors using integer trial division. Independent integer Horner evaluation at every one of the 7 585 divisors above b gives no zero. Thus neither endpoint vanishes for any integer $c > b$, with no imposed upper bound on c . Theorem 12.2 supplies a positive root in $(0, 3)$; it can equal neither one nor two, proving nonintegrality. $\square$

The reflected quotient in this diagnostic uses $N = |V(H)| = a + b + c + ab + ac + bc$, not the full graph vertex count. The inclusive requested bound $b \leq 15$ and exact rational linear-factor extraction are cross-checked against the independent divisor certificate. The empty lists settle this pair range; they do not prove either endpoint surface globally empty.

Proposition C.6. *Every positive three-prime exponent triple congruent to $(2, 2, 2)$, or to a permutation of $(1, 2, 2)$, modulo three gives a nonintegral ideal intersection graph.*

Proof. For these two residue patterns, reduction of the general complement quintic gives, respectively,

$$h_C(x) \equiv x^5, \quad h_C(x) \equiv (x^2 + 1)(x^3 + x^2 - x + 1) \pmod{3}.$$

The corresponding pairs $(h_C(1), h_C(2))$ are $(1, 2)$ and $(1, 1)$ modulo three, so neither integer endpoint is a root. Minimum exponents at most three are already covered. Otherwise Theorem 12.2 supplies a positive root in $(0, 3)$, which cannot be an integer. Its graph lift is noninteger as well. For the second pattern, the irreducible factor $x^2 + 1$ also directly rules out a completely integral quotient spectrum. $\square$

For example, $(8, 17, 22)$ and $(8, 17, 23)$ satisfy the new conditions with gcd one and unequal 2-adic valuations. Complete endpoint tables modulo two and three, checked against independent integer determinants, are given in the residue certificate. Every pair (a, b) modulo either prime has at least one c root on each surface; retaining the full root sets, rather than only root-free pairs, is necessary to detect these exclusions.

Remark C.7. Endpoint tests at a fixed finite set of moduli cannot by themselves prove either surface empty on all fully distinct positive triples. Let M be their least common multiple. Since $(9, 9, 136)$ has $h_C(1) = 0$, the strictly ordered triples $(9 + kM, 9 + (k + 1)M, 136 + (k + 2)M)$ pass every selected endpoint-one test. The known endpoint-two zero $(10, 10, 12)$ similarly gives survivors $(10 + kM, 10 + (k + 1)M, 12 + (k + 2)M)$. For large k their middle exponents exceed fifteen. These are congruence survivors, not asserted integer zeros. These fixed-span tails are excluded by Corollary 12.4. To retain the geometric restrictions, take L divisible by M and eight and any $d \geq 32$ divisible by L . The triples $(9 + d, 9 + 2d, 136 + 4d)$ and $(10 + d, 10 + 2d, 12 + 4d)$ pass the respective endpoint tests, have gcd one and two, and satisfy both upper bounds on the maximum while failing the sufficient inertia inequality. The substitution identities are given in the accompanying scope note. This concerns endpoint-value tests alone; another-root obstruction or further arithmetic information can still exclude these candidates.

The remaining all-even triples have gcd two, $8 \leq a < b < c < R(a)$ and $b \geq 16$, outside the preceding proved regions. The mixed-parity surface $h_C(1) = 0$ still needs another obstruction.